Online Appendix to Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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A Graduate School Regression Results

The main dataset in this paper consists of all 2,329 applicants to the university's economics PhD program from 2015–2019. However, I also have another dataset consisting of all 6,222 applicants to the university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023. Unlike the economics department dataset, the graduate school dataset does not contain subjective variables or program outcomes data. Therefore, I estimate this paper's structural models on the economics department dataset only. However, the graduate school dataset, which still contains objective variables and each applicant's admission status, is necessary for making comparisons across programs. The graduate school dataset also contains an ethnicity variable and concentration variables for the non-economics programs. Compared with fields of interest in economics, concentrations are more binding because applicants cannot costlessly change their concentration after entering the program. Summary statistics and correlations for the graduate school data are in Tables 1 and 2; the economics program has the highest acceptance rate at 11.21%, whereas the government program has the lowest acceptance rate at 4.30%.

In Table 3, I present logistic regressions of acceptance on the objective variables for all five programs.[1] The left subtable contains the regression coefficients, whereas the right subtable contains the average marginal effects (AMEs). Relative to the other programs, economics places more weight on undergraduate GPA and the quantitative GRE. A one point increase in an applicant's quantitative GRE percentile increases their acceptance probability by over one percentage point. As Bai, Esche, MacLeod, and Shi (2022) find, having a prolific recommender, defined as one who also recommended other applicants in the sample, matters as well. Going from zero to three prolific recommenders increases an applicant's acceptance probability by over six percentage points. This result provides evidence of academic network effects in the social sciences.

There are a few more results worth describing. First, across programs, acceptance rates were lowest in Spring 2021, which was the year that Covid reduced department funding. Second, all else equal, economics favors younger applicants, and economics, government, and history favor women. Third, government and history favor underrepresented minorities, whereas economics disfavors applicants from East and South Asia. Fourth, relative to a non-elite U.S. undergraduate institution, an elite institution matters more than a graduate degree or work experience. Fifth, taking a high number of math courses matters for economics. Note, however, that these results are descriptive and by themselves do not support any normative claims about admissions. In addition, the Pseudo-R^2 values never exceed 0.3 for any program. However, they are highest for economics at 0.2677, meaning that economics admissions is more data-driven than the other programs, at least with respect to the observable variables. That said, some of the other programs have admissions criteria, such as interviews, that economics does not. Although I cannot observe

[1] In Table 4, I present the regression results on synthetic data. The synthetic estimates are more often than random chance similar to the actual estimates, but the synthetic estimates are generally less statistically significant.

interview results, interviews may explain some of the variation in acceptances in those programs.

I also run lasso with 5-fold cross-validation for variable selection. Per Table 5, lasso selects more variables for economics than for any other program. The last three variables in this table are missing variable dummies. Any time an applicant's GRE percentiles, undergraduate GPA, or TOEFL/IELTS scores are missing, I impute the variable's sample mean for that observation and encode the missing variable dummy equal to 1 (Krueger & Wu 2000). Allowing lasso to select one but not the other (see Bai et al. 2022) has a meaningful interpretation. For example, when lasso selects GPA but not its missing dummy for economics, it means that GPAs, as opposed to TOEFL/IELTS scores where lasso selects the missing dummy, are missing at random.

Finally, I implement a Wald test of whether the determinants of acceptance are the same across programs. The test's setup is as follows. Because the samples across programs are different, I assume that $\left(Y^E \perp\!\!\!\perp Y^{NE}\right)\big|X$; that is, conditional on the independent variables X, the dependent variable for economics Y^E is independent of its non-economics counterpart Y^{NE}. Also, suppose the dependent variable is binary; for example, accepted vs. rejected. As such, I parameterize the economics and non-economics outcome probabilities with the logistic functional form: $P\left(Y_n^E = 1\big|x_n^E;\theta^E\right) = \frac{\exp(x_{n,0}^E\theta_0^E + x_{n,1}^E\theta_1^E)}{1+\exp(x_{n,0}^E\theta_0^E + x_{n,1}^E\theta_1^E)}$ and $P\left(Y_n^{NE} = 1\big|x_n^{NE};\theta^{NE}\right) = \frac{\exp(x_{n,0}^{NE}\theta_0^{NE} + x_{n,1}^{NE}\theta_1^{NE})}{1+\exp(x_{n,0}^{NE}\theta_0^{NE} + x_{n,1}^{NE}\theta_1^{NE})}$.

Notably, I separate the intercept θ_0 from the slope θ_1, since the goal is to test for equal slopes, not equal intercepts. That said, I can add slopes I do not test to θ_0. Then the likelihood is:

$$\begin{aligned}\mathcal{L}_N^U\left(y\big|x;\theta\right) = &\prod_{n=1}^{N_E} P\left(Y_n^E = 1\big|x_n^E;\theta^E\right)^{Y_n^E}\left[1 - P\left(Y_n^E = 1\big|x_n^E;\theta^E\right)\right]^{1-Y_n^E} \\ &\prod_{n=1}^{N_{NE}} P\left(Y_n^{NE} = 1\big|x_n^{NE};\theta^{NE}\right)^{Y_n^{NE}}\left[1 - P\left(Y_n^{NE} = 1\big|x_n^{NE};\theta^{NE}\right)\right]^{1-Y_n^{NE}}.\end{aligned} \tag{1}$$

Let $g(\hat{\theta}) = \frac{1}{N_E}\sum_{n=1}^{N_E}\hat{P}_n^E(1-\hat{P}_n^E)\hat{\theta}_1^E - \frac{1}{N_{NE}}\sum_{n=1}^{N_{NE}}\hat{P}_n^{NE}(1-\hat{P}_n^{NE})\hat{\theta}_1^{NE}$. The Wald test statistic is therefore $\mathrm{W} = g(\hat{\theta})'\left[\nabla_\theta g(\hat{\theta})\hat{V}(\hat{\theta})\nabla_\theta g(\hat{\theta})'\right]^{-1} g(\hat{\theta}) \sim \chi_Q^2$, where Q is the number of restrictions, i.e. the number of determinants I test at a time. Note that the null hypothesis is of equal AMEs, not equal parameter estimates, since only AMEs can be compared across samples (Mood 2010).

If the dependent variable were continuous (e.g. placement rank), I would parameterize its economics and non-economics distributions as: $f\left(y_n^E\big|x_n^E;\beta^E,\sigma_E^2\right) = \frac{1}{\sqrt{2\pi\sigma_E^2}}\exp\left[\frac{-1}{2\sigma_E^2}\left(y_n^E - x_{n,0}^E\beta_0^E - x_{n,1}^E\beta_1^E\right)^2\right]$ and $f\left(y_n^{NE}\big|x_n^{NE};\beta^{NE},\sigma_{NE}^2\right) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}}\exp\left[\frac{-1}{2\sigma_{NE}^2}\left(y_n^{NE} - x_{n,0}^{NE}\beta_0^{NE} - x_{n,1}^{NE}\beta_1^{NE}\right)^2\right]$. Then:

$$\mathcal{L}_N^U\left(y\big|x;\beta,\sigma^2\right) = \prod_{n=1}^{N_E} f\left(y_n^E\big|x_n^E;\beta^E,\sigma_E^2\right)\prod_{n=1}^{N_{NE}} f\left(y_n^{NE}\big|x_n^{NE};\beta^{NE},\sigma_{NE}^2\right). \tag{2}$$

Since linear coefficients can be compared across samples, the test is simpler (and can also be done with a likelihood-ratio test). Let $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$. Under homoskedasticity (i.e. $\forall n, \sigma_n^2 = \sigma^2$), the Wald statistic is $\mathrm{W} = g(\hat{\beta}_1)'\left[\nabla_{\beta,\sigma^2} g(\hat{\beta}_1)\hat{V}(\hat{\beta},\hat{\sigma}^2)\nabla_{\beta,\sigma^2} g(\hat{\beta}_1)'\right]^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$.

Returning to Table 3b, the first column contains the economics acceptance AMEs, the sixth column contains the non-economics acceptance AMEs, and the Wald test resoundingly rejects

the null hypothesis of equal AMEs with $p < 0.0001$. Therefore, the differences in AMEs for economics vs. non-economics are highly significant. To ensure that this test does not over-reject equal AMEs in general, I simulate two datasets, which for simplicity are for only the economics program and two non-economics programs. In each dataset, I simulate acceptance and success dependent variables. In the first dataset, the means, standard deviations, and correlations are similar to the actual data, meaning that the determinants of acceptance and success are different across programs.[2] In the second dataset, where the test is not supposed to reject equal AMEs, I modify the correlations, so the non-economics determinants are the same as those for economics.

In Table 6, I present Wald test results on the simulated data. In the dataset where the data-generating parameters are such that the true AMEs of the determinants of acceptance and success are the same, the test does not reject the null of equal determinants; I obtain $p = 0.6843$ for the acceptance dependent variable and $p = 0.6993$ for the success dependent variable. However, in the dataset where the determinants are different, the test rejects with $p < 0.0001$. Therefore, on the simulated data, the Wald test works properly, rejecting resoundingly when it should and not rejecting when it should not. Because the simulated dataset has 5,000 observations, which is comparable to the number of observations in the actual data, the test is sufficiently powerful.

As I explain in Section 3, I cannot use a Wald test to compare the determinants of acceptance to those of success within programs as a test of optimality. However, I can answer additional questions within this framework, as long as those questions are not about optimality. For example, I can test if there are different determinants of different types of success. Specifically, applicants may need different skills at different stages of the program; Athey, Katz, Krueger, Levitt, and Poterba (2007) find that the quantitative GRE matters for course grades, whereas research experience matters more for placements. Testing this hypothesis may be useful for committees who must weigh the early vs. late stages when deciding which applicants to accept. I can also test if the determinants of admission or success are different for U.S. vs. international applicants (see Krueger and Wu 2000). International applicants may have different levels of preparation or research interests, and be more likely to complete the program for lack of outside options and return to their native country after graduation. Finally, I can test if the determinants have changed over time. Reasons for such changes may include Covid and other macro/geopolitical factors, and for economics in particular, a shift toward elite predocs and Federal Reserve RAships.

[2]For some programs, the initial post-program job placement matters less career-wise than it does for economics, which is worth considering when interpreting Wald test results for success across programs.

Table 1: Graduate School Means by Program

(a) All Applicants

	Economics	Government	History	Linguistics	Psychology
Accept	0.1121	0.0430	0.0986	0.0678	0.0535
Covid Year	0.2478	0.2792	0.3029	0.3460	0.3004
Winsorized Age	26.2782	27.9633	26.8478	28.0692	25.8230
Female	0.4201	0.4182	0.4232	0.6133	0.7490
U.S. African/Hispanic/Native	0.0289	0.0766	0.1159	0.0665	0.1728
U.S. Asian	0.0313	0.0421	0.0304	0.0353	0.0576
International Asian	0.5322	0.2510	0.1058	0.2714	0.1523
International Other	0.2517	0.2967	0.2203	0.3256	0.1584
GRE Verbal Percentile	71.8677	81.3486	82.6054	73.5451	75.6192
GRE Quantitative Percentile	86.2163	63.3428	49.3295	57.8455	58.0000
GRE Analytic Percentile	52.6351	72.1905	75.0460	63.6824	69.2077
Undergraduate GPA	3.6534	3.5675	3.6531	3.6567	3.5695
TOEFL/IELTS	7.5232	7.5422	7.4252	7.5444	7.3158
Elite U.S. University	0.1516	0.1847	0.2565	0.1954	0.2840
Elite U.S. Liberal Arts College	0.0423	0.0672	0.0696	0.0231	0.0535
Elite Foreign University	0.0717	0.0578	0.0478	0.0488	0.0206
Other Foreign University	0.5736	0.3953	0.2333	0.5102	0.1852
Graduate Degree	0.7478	0.7553	0.6652	0.7544	0.4259
Work Experience	0.5938	0.7463	0.6638	0.7544	0.7593
#Professor Recommenders	2.5173	2.3469	2.6261	2.4355	2.1831
#Prolific Recommenders	1.0760	0.6912	0.5043	0.7069	0.3128
#Math Courses	6.6396				
Advanced Math	0.4827				
Missing GRE	0.0255	0.4612	0.6217	0.6839	0.4650
Missing Undergraduate GPA	0.6607	0.4827	0.3072	0.5726	0.2490
Missing TOEFL/IELTS	0.5958	0.7714	0.8449	0.7707	0.8827
Observations	2078	2231	690	737	486

(b) Accept = 1

	Economics	Government	History	Linguistics	Psychology
Covid Year	0.1888	0.2188	0.2059	0.2200	0.2308
Winsorized Age	25.0815	27.7292	27.2794	27.1000	25.7692
Female	0.4850	0.5000	0.5588	0.5800	0.6154
U.S. African/Hispanic/Native	0.0386	0.1458	0.1765	0.0400	0.0769
U.S. Asian	0.0429	0.0312	0.0147	0.0600	0.1154
International Asian	0.4163	0.1667	0.0882	0.3600	0.0769
International Other	0.2575	0.2292	0.3235	0.2000	0.1538
GRE Verbal Percentile	84.3980	88.8725	86.6328	77.9161	85.7729
GRE Quantitative Percentile	91.9289	68.3708	54.8113	64.5504	62.3462
GRE Analytic Percentile	69.0355	82.7222	78.0879	68.8958	74.6169
Undergraduate GPA	3.7336	3.6404	3.7204	3.7004	3.6341
TOEFL/IELTS	7.6419	7.6050	7.4730	7.6237	7.3178
Elite U.S. University	0.2833	0.3125	0.3382	0.2200	0.5385
Elite U.S. Liberal Arts College	0.1030	0.1458	0.1176	0.0200	0.1154
Elite Foreign University	0.1030	0.0521	0.0588	0.1000	0.0385
Other Foreign University	0.3863	0.2396	0.2647	0.4000	0.1154
Graduate Degree	0.5837	0.6771	0.7206	0.7400	0.4615
Work Experience	0.6309	0.8125	0.7647	0.7000	0.9231
#Professor Recommenders	2.5494	2.3750	2.8235	2.7200	2.2308
#Prolific Recommenders	1.2833	0.9167	0.6765	1.0800	0.4615
#Math Courses	7.2575				
Advanced Math	0.6781				
Missing GRE	0.0086	0.2917	0.6324	0.5800	0.1923
Missing Undergraduate GPA	0.4893	0.3125	0.3088	0.4800	0.1923
Missing TOEFL/IELTS	0.6567	0.8854	0.8529	0.7600	0.8846
Observations	233	96	68	50	26

Table 2: Graduate School Correlation Matrix

	Covid	Age	Fem	USAHN	USAsn	IntAsn	IntOth	GREV	GREQ	GREA	GPA	T/I	EUSUni	EUSLA	EFor	OthFor	Grad	Work	Prof	Pro	MGRE	MGPA	MT/I
Covid Year	1.00																						
Winsorized Age	0.01	1.00																					
Female	0.02	-0.13	1.00																				
U.S. African/Hispanic/Native	0.00	-0.04	0.04	1.00																			
U.S. Asian	-0.00	-0.06	0.05	-0.05	1.00																		
International Asian	-0.05	-0.07	0.02	-0.19	-0.14	1.00																	
International Other	0.01	0.18	-0.06	-0.17	-0.12	-0.42	1.00																
GRE Verbal Percentile	0.03	-0.06	-0.04	0.00	0.04	-0.12	-0.15	1.00															
GRE Quantitative Percentile	-0.02	-0.16	-0.10	-0.18	0.00	0.44	-0.10	0.16	1.00														
GRE Analytic Percentile	0.05	-0.06	0.03	0.08	0.08	-0.37	-0.08	0.60	-0.20	1.00													
Undergraduate GPA	-0.02	-0.29	0.07	-0.13	-0.02	0.02	0.00	0.06	0.12	0.05	1.00												
TOEFL/IELTS	-0.01	-0.07	0.01	-0.02	0.01	0.04	0.00	0.24	0.12	0.19	0.00	1.00											
Elite U.S. University	0.01	-0.12	0.02	0.12	0.14	-0.12	-0.22	0.16	-0.00	0.19	-0.01	-0.02	1.00										
Elite U.S. Liberal Arts College	-0.02	-0.02	0.04	0.04	0.04	-0.08	-0.09	0.12	-0.01	0.13	-0.02	-0.00	-0.11	1.00									
Elite Foreign University	-0.00	-0.04	-0.00	-0.06	-0.04	0.09	0.09	0.02	0.08	-0.02	0.01	0.04	-0.12	-0.06	1.00								
Other Foreign University	-0.02	0.20	-0.04	-0.23	-0.14	0.34	0.39	-0.26	0.20	-0.39	0.02	0.02	-0.43	-0.21	-0.22	1.00							
Graduate Degree	0.00	0.41	-0.08	-0.13	-0.09	0.21	0.15	-0.14	0.05	-0.20	-0.22	0.03	-0.21	-0.10	0.05	0.37	1.00						
Work Experience	0.01	0.33	0.05	0.05	0.01	-0.18	0.11	0.01	-0.15	0.11	-0.08	-0.03	-0.02	0.03	-0.03	-0.01	0.10	1.00					
#Professor Recommenders	-0.02	-0.19	-0.04	-0.04	0.01	0.11	-0.09	0.01	0.09	-0.04	0.14	0.03	-0.02	0.02	-0.00	-0.02	-0.03	-0.18	1.00				
#Prolific Recommenders	0.01	-0.06	-0.02	-0.07	-0.01	0.24	-0.10	-0.01	0.25	-0.11	0.04	0.10	-0.01	-0.01	0.06	0.08	0.17	-0.09	0.19	1.00			
Missing GRE	0.09	0.15	0.05	0.11	-0.01	-0.22	0.17	0.09	-0.33	0.17	-0.09	-0.10	-0.03	-0.04	-0.03	-0.01	0.02	0.10	-0.08	-0.20	1.00		
Missing Undergraduate GPA	-0.01	0.18	-0.04	-0.24	-0.15	0.36	0.42	-0.24	0.21	-0.38	0.02	0.04	-0.46	-0.23	0.23	0.83	0.37	-0.02	-0.03	0.10	-0.01	1.00	
Missing TOEFL/IELTS	0.05	-0.07	0.03	0.16	0.11	-0.32	-0.22	0.16	-0.24	0.32	-0.02	-0.02	0.29	0.14	-0.04	-0.58	-0.20	0.05	-0.07	-0.05	0.12	-0.58	1.00
Observations	6222																						

Table 3: Graduate School Logistic Regression Results

(a) Coefficients

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3966**	-0.4332*	-0.5752*	-0.5501	-0.1766
	(0.2010)	(0.2612)	(0.3479)	(0.4129)	(0.5545)
Winsorized Age	-0.0995***	0.0454*	0.0380	0.0222	-0.0520
	(0.0358)	(0.0271)	(0.0327)	(0.0532)	(0.0899)
Female	0.4079**	0.6723***	0.6539**	0.1311	-0.5060
	(0.1681)	(0.2299)	(0.2971)	(0.3452)	(0.5606)
U.S. African/Hispanic/Native	0.4221	1.0081***	1.1601***	0.0120	-0.8263
	(0.5258)	(0.3545)	(0.4426)	(0.8411)	(1.1611)
U.S. Asian	-0.5394	-0.5669	-0.3260	0.8638	1.1711
	(0.4199)	(0.6504)	(1.2512)	(0.7201)	(0.7276)
International Asian	-0.8445***	-0.1853	0.9405	0.4890	-0.7827
	(0.2945)	(0.4146)	(0.7233)	(0.6729)	(0.9323)
International Other	0.5034	0.3781	1.3523***	-0.1061	0.6239
	(0.3172)	(0.4131)	(0.5086)	(0.7483)	(1.1070)
GRE Verbal Percentile	0.0155***	0.0281*	0.0753**	-0.0092	0.0607***
	(0.0060)	(0.0160)	(0.0344)	(0.0154)	(0.0218)
GRE Quantitative Percentile	0.1547***	0.0161*	0.0210*	0.0211	-0.0138
	(0.0186)	(0.0086)	(0.0116)	(0.0150)	(0.0159)
GRE Analytic Percentile	0.0179***	0.0199**	0.0030	0.0175	-0.0202
	(0.0042)	(0.0094)	(0.0134)	(0.0127)	(0.0218)
Undergraduate GPA	2.2212***	0.6968	1.3648**	1.2645	1.7772*
	(0.7219)	(0.5476)	(0.5513)	(0.9703)	(0.9487)
TOEFL/IELTS	0.3878	0.6456	1.0963**	0.9655*	-0.4656
	(0.2883)	(0.4708)	(0.5526)	(0.5456)	(0.6112)
Elite U.S. University	0.8850***	0.3943	0.6541*	-0.6057	1.4303**
	(0.2960)	(0.3135)	(0.3802)	(0.4985)	(0.6192)
Elite U.S. Liberal Arts	1.1173***	0.6276*	1.0698**	-0.6088	2.3151**
	(0.3813)	(0.3754)	(0.5399)	(1.4186)	(1.0642)
Elite Foreign University	1.0061*	0.4759	1.7715**	1.6195*	0.3889
	(0.6060)	(0.8224)	(0.7938)	(0.8587)	(1.9889)
Other Foreign University	0.5442	0.5777	1.3733**	0.4188	-1.0737
	(0.5765)	(0.7686)	(0.6992)	(0.7157)	(1.3016)
Graduate Degree	0.0467	0.0733	0.3128	0.4049	0.4892
	(0.2238)	(0.2583)	(0.3280)	(0.4896)	(0.6399)
Work Experience	0.3932**	0.4027	0.4972	0.1495	1.8201**
	(0.1889)	(0.2993)	(0.3295)	(0.3778)	(0.8564)
#Professor Recommenders	0.0340	-0.0179	0.5511**	0.4781*	-0.0981
	(0.1317)	(0.1339)	(0.2554)	(0.2662)	(0.2660)
#Prolific Recommenders	0.2635***	0.3025**	0.1880	0.3284**	0.5901*
	(0.0854)	(0.1262)	(0.1576)	(0.1544)	(0.3145)
#Math Courses	0.1689***				
	(0.0596)				
Advanced Math	0.2319				
	(0.1790)				
Missing GRE	0.1899	-0.3259	0.6282	-0.0831	-1.1514**
	(0.7646)	(0.3100)	(0.4275)	(0.3761)	(0.5702)
Missing Undergraduate GPA	-0.4745	-0.4917	-1.5492**	-1.1097*	0.4690
	(0.4891)	(0.5440)	(0.6893)	(0.6481)	(1.1831)
MissingTOEFL/IELTS	-0.3786	0.6188	0.6745	-0.0440	-1.7224*
	(0.2432)	(0.5657)	(0.5190)	(0.5099)	(0.9271)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.2677	0.1525	0.1982	0.1871	0.2475

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) AMEs

Depdendent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0303**	-0.0164	-0.0431*	-0.0298	-0.0075	-0.0218***
	(0.0153)	(0.0100)	(0.0261)	(0.0224)	(0.0238)	(0.0084)
Winsorized Age	-0.0076***	0.0017*	0.0028	0.0012	-0.0022	0.0016*
	(0.0027)	(0.0010)	(0.0025)	(0.0029)	(0.0038)	(0.0009)
Female	0.0312**	0.0254***	0.0490**	0.0071	-0.0216	0.0207***
	(0.0129)	(0.0088)	(0.0220)	(0.0187)	(0.0239)	(0.0075)
U.S. African/Hispanic/Native	0.0323	0.0381***	0.0869***	0.0007	-0.0352	0.0384***
	(0.0402)	(0.0137)	(0.0332)	(0.0455)	(0.0487)	(0.0119)
U.S. Asian	-0.0413	-0.0215	-0.0244	0.0468	0.0499	0.0062
	(0.0322)	(0.0247)	(0.0936)	(0.0391)	(0.0313)	(0.0185)
International Asian	-0.0646***	-0.0070	0.0704	0.0265	-0.0334	0.0069
	(0.0224)	(0.0157)	(0.0543)	(0.0363)	(0.0393)	(0.0132)
International Other	0.0385	0.0143	0.1013***	-0.0057	0.0266	0.0238*
	(0.0243)	(0.0157)	(0.0385)	(0.0406)	(0.0471)	(0.0132)
GRE Verbal Percentile	0.0012***	0.0011*	0.0056**	-0.0005	0.0026***	0.0015***
	(0.0005)	(0.0006)	(0.0026)	(0.0008)	(0.0010)	(0.0005)
GRE Quantitative Percentile	0.0118***	0.0006*	0.0016*	0.0011	-0.0006	0.0006**
	(0.0014)	(0.0003)	(0.0009)	(0.0008)	(0.0007)	(0.0003)
GRE Analytic Percentile	0.0014***	0.0008**	0.0002	0.0009	-0.0009	0.0007**
	(0.0003)	(0.0004)	(0.0010)	(0.0007)	(0.0009)	(0.0003)
Undergraduate GPA	0.1699***	0.0264	0.1022**	0.0684	0.0757*	0.0488***
	(0.0551)	(0.0208)	(0.0423)	(0.0522)	(0.0418)	(0.0176)
TOEFL/IELTS	0.0297	0.0244	0.0821**	0.0523*	-0.0198	0.0326***
	(0.0219)	(0.0179)	(0.0417)	(0.0301)	(0.0259)	(0.0123)
Elite U.S. University	0.0677***	0.0149	0.0490*	-0.0328	0.0610**	0.0233**
	(0.0225)	(0.0119)	(0.0281)	(0.0264)	(0.0271)	(0.0098)
Elite U.S. Liberal Arts College	0.0855***	0.0238*	0.0801**	-0.0330	0.0987**	0.0420***
	(0.0288)	(0.0143)	(0.0402)	(0.0761)	(0.0480)	(0.0136)
Elite Foreign University	0.0770*	0.0180	0.1327**	0.0877*	0.0166	0.0479**
	(0.0463)	(0.0311)	(0.0591)	(0.0481)	(0.0850)	(0.0241)
Other Foreign University	0.0416	0.0219	0.1028*	0.0227	-0.0458	0.0273
	(0.0441)	(0.0291)	(0.0527)	(0.0391)	(0.0556)	(0.0221)
Graduate Degree	0.0036	0.0028	0.0234	0.0219	0.0208	0.0106
	(0.0171)	(0.0098)	(0.0247)	(0.0262)	(0.0273)	(0.0089)
Work Experience	0.0301**	0.0152	0.0372	0.0081	0.0776**	0.0221**
	(0.0144)	(0.0113)	(0.0248)	(0.0205)	(0.0368)	(0.0089)
#Professor Recommenders	0.0026	-0.0007	0.0413**	0.0259*	-0.0042	0.0063
	(0.0101)	(0.0051)	(0.0190)	(0.0142)	(0.0113)	(0.0047)
#Prolific Recommenders	0.0202***	0.0114**	0.0141	0.0178**	0.0251*	0.0133***
	(0.0065)	(0.0048)	(0.0118)	(0.0082)	(0.0130)	(0.0038)
#Math Courses	0.0129***					
	(0.0045)					
Advanced Math	0.0177					
	(0.0137)					
Missing GRE	0.0145	-0.0123	0.0470	-0.0045	-0.0491*	-0.0062
	(0.0585)	(0.0118)	(0.0319)	(0.0204)	(0.0252)	(0.0088)
Missing Undergraduate GPA	-0.0363	-0.0186	-0.1160**	-0.0601*	0.0200	-0.0291
	(0.0375)	(0.0206)	(0.0524)	(0.0355)	(0.0505)	(0.0179)
Missing TOEFL/IELTS	-0.0290	0.0234	0.0505	-0.0024	-0.0734*	0.0108
	(0.0188)	(0.0215)	(0.0388)	(0.0276)	(0.0417)	(0.0135)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic | DF | P-Value	115.8837	23	0.0000			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 4: Graduate School Logistic Regression Results (Synthetic)

(a) Coefficients

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.1997	-0.0323	0.2028	-0.4791	-0.5946
	(0.1633)	(0.2215)	(0.3078)	(0.3477)	(0.4520)
Winsorized Age	-0.0509**	-0.0198	0.0576*	-0.0164	0.1203***
	(0.0201)	(0.0174)	(0.0307)	(0.0357)	(0.0400)
Female	0.1941	0.0923	0.1389	0.2595	0.2119
	(0.1402)	(0.2004)	(0.2526)	(0.3255)	(0.4310)
U.S. African/Hispanic/Native	0.3053	-0.1457	-0.2379	-1.6401	0.5413
	(0.3724)	(0.4303)	(0.4670)	(1.0720)	(0.5306)
U.S. Asian	0.5663	0.6016	-0.7934	-0.4305	0.7122
	(0.3642)	(0.4452)	(1.1675)	(0.7936)	(0.8863)
International Asian	-0.1428	0.1630	0.5780	-0.4965	0.1868
	(0.2169)	(0.2803)	(0.3740)	(0.4275)	(0.6866)
International Other	-0.1231	0.1395	0.1097	-0.1602	-1.0807
	(0.2311)	(0.3046)	(0.3237)	(0.3832)	(0.7583)
GRE Verbal Percentile	0.0167***	0.0247**	0.0164	-0.0029	0.0132
	(0.0048)	(0.0117)	(0.0127)	(0.0100)	(0.0184)
GRE Quantitative Percentile	0.0233***	-0.0027	0.0026	0.0304**	0.0097
	(0.0090)	(0.0051)	(0.0088)	(0.0137)	(0.0099)
GRE Analytic Percentile	0.0042	0.0004	-0.0073	-0.0042	0.0069
	(0.0031)	(0.0059)	(0.0092)	(0.0101)	(0.0103)
Undergraduate GPA	0.2292	0.6327	1.3552**	-1.3171**	-0.1002
	(0.4745)	(0.4985)	(0.5796)	(0.5584)	(0.6152)
TOEFL/IELTS	-0.4673**	0.5141	-0.7648	0.2446	-0.7354
	(0.2157)	(0.4185)	(0.6814)	(0.7498)	(0.8025)
Elite U.S. University	-0.4850*	-0.2243	0.0572	0.4764	0.3941
	(0.2560)	(0.3258)	(0.3421)	(0.4788)	(0.4782)
Elite U.S. Liberal Arts College	0.2429	0.1759	0.2358	0.4860	0.8652
	(0.3049)	(0.4042)	(0.4964)	(0.9631)	(0.7261)
Elite Foreign University	-0.1622	-1.1929	0.7404	0.4801	3.0530***
	(0.3015)	(0.7484)	(0.5668)	(0.7841)	(0.8673)
Other Foreign University	-0.1087	0.2043	0.2721	0.1731	0.3023
	(0.1979)	(0.2732)	(0.3391)	(0.4292)	(0.5926)
Graduate Degree	-0.0198	0.1770	-0.1601	-0.8698***	-0.2769
	(0.1624)	(0.2439)	(0.2765)	(0.3320)	(0.3814)
Work Experience	0.2525*	-0.0391	-0.0736	-0.2650	-1.1936**
	(0.1441)	(0.2396)	(0.2629)	(0.3417)	(0.4805)
#Professor Recommenders	-0.1059	0.0911	-0.1828	0.2181	-0.4923**
	(0.0909)	(0.1138)	(0.1473)	(0.1817)	(0.1956)
#Prolific Recommenders	0.0039	0.2189**	-0.1743	0.0673	0.5134**
	(0.0656)	(0.1108)	(0.1723)	(0.1642)	(0.2213)
#Math Courses	0.0366				
	(0.0421)				
Advanced Math	0.1961				
	(0.1416)				
Missing GRE	-0.1993	-0.2029	-0.0011	-0.0433	-0.6040
	(0.4328)	(0.2195)	(0.2689)	(0.3503)	(0.4017)
Missing Undergraduate GPA	0.1531	0.0895	-0.2455	0.6298**	-0.3574
	(0.1548)	(0.2291)	(0.3015)	(0.3125)	(0.4342)
Missing TOEFL/IELTS	-0.2950*	0.4898	0.5752	0.1346	-2.0781***
	(0.1640)	(0.3480)	(0.5466)	(0.4345)	(0.7633)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.0524	0.0366	0.0753	0.1046	0.2230

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) AMEs

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0204	-0.0014	0.0192	-0.0298	-0.0345	-0.0089
	(0.0167)	(0.0099)	(0.0291)	(0.0216)	(0.0262)	(0.0086)
Winsorized Age	-0.0052**	-0.0009	0.0054*	-0.0010	0.0070***	0.0004
	(0.0021)	(0.0008)	(0.0029)	(0.0022)	(0.0023)	(0.0008)
Female	0.0198	0.0041	0.0131	0.0162	0.0123	0.0099
	(0.0143)	(0.0089)	(0.0239)	(0.0204)	(0.0251)	(0.0078)
U.S. African/Hispanic/Native	0.0312	-0.0065	-0.0225	-0.1021	0.0314	-0.0137
	(0.0381)	(0.0192)	(0.0441)	(0.0673)	(0.0306)	(0.0147)
U.S. Asian	0.0579	0.0268	-0.0750	-0.0268	0.0413	0.0241
	(0.0372)	(0.0199)	(0.1101)	(0.0495)	(0.0516)	(0.0179)
International Asian	-0.0146	0.0073	0.0546	-0.0309	0.0108	0.0075
	(0.0222)	(0.0125)	(0.0353)	(0.0263)	(0.0399)	(0.0112)
International Other	-0.0126	0.0062	0.0104	-0.0100	-0.0627	-0.0014
	(0.0236)	(0.0136)	(0.0306)	(0.0237)	(0.0443)	(0.0106)
GRE Verbal Percentile	0.0017***	0.0011**	0.0015	-0.0002	0.0008	0.0008**
	(0.0005)	(0.0005)	(0.0012)	(0.0006)	(0.0010)	(0.0004)
GRE Quantitative Percentile	0.0024***	-0.0001	0.0003	0.0019**	0.0006	0.0003
	(0.0009)	(0.0002)	(0.0008)	(0.0009)	(0.0006)	(0.0002)
GRE Analytic Percentile	0.0004	0.0000	-0.0007	-0.0003	0.0004	0.0000
	(0.0003)	(0.0003)	(0.0009)	(0.0006)	(0.0006)	(0.0002)
Undergraduate GPA	0.0234	0.0282	0.1281**	-0.0820**	-0.0058	0.0130
	(0.0485)	(0.0223)	(0.0553)	(0.0358)	(0.0357)	(0.0158)
TOEFL/IELTS	-0.0478**	0.0229	-0.0723	0.0152	-0.0427	0.0050
	(0.0221)	(0.0187)	(0.0645)	(0.0466)	(0.0469)	(0.0187)
Elite U.S. University	-0.0496*	-0.0100	0.0054	0.0297	0.0229	0.0036
	(0.0261)	(0.0145)	(0.0324)	(0.0297)	(0.0274)	(0.0108)
Elite U.S. Liberal Arts College	0.0248	0.0078	0.0223	0.0303	0.0502	0.0217
	(0.0312)	(0.0180)	(0.0469)	(0.0602)	(0.0418)	(0.0160)
Elite Foreign University	-0.0166	-0.0532	0.0700	0.0299	0.1771***	0.0158
	(0.0308)	(0.0337)	(0.0537)	(0.0490)	(0.0484)	(0.0179)
Other Foreign University	-0.0111	0.0091	0.0257	0.0108	0.0175	0.0116
	(0.0202)	(0.0122)	(0.0321)	(0.0267)	(0.0344)	(0.0103)
Graduate Degree	-0.0020	0.0079	-0.0151	-0.0542***	-0.0161	-0.0093
	(0.0166)	(0.0109)	(0.0261)	(0.0206)	(0.0220)	(0.0085)
Work Experience	0.0258*	-0.0017	-0.0070	-0.0165	-0.0692**	-0.0093
	(0.0148)	(0.0107)	(0.0248)	(0.0213)	(0.0275)	(0.0086)
#Professor Recommenders	-0.0108	0.0041	-0.0173	0.0136	-0.0286***	-0.0027
	(0.0093)	(0.0051)	(0.0140)	(0.0114)	(0.0110)	(0.0042)
#Prolific Recommenders	0.0004	0.0098*	-0.0165	0.0042	0.0298**	0.0083*
	(0.0067)	(0.0050)	(0.0164)	(0.0102)	(0.0128)	(0.0045)
#Math Courses	0.0037					
	(0.0043)					
Advanced Math	0.0201					
	(0.0145)					
Missing GRE	-0.0204	-0.0090	-0.0001	-0.0027	-0.0350	-0.0098
	(0.0443)	(0.0098)	(0.0254)	(0.0218)	(0.0239)	(0.0079)
Missing Undergraduate GPA	0.0157	0.0040	-0.0232	0.0392*	-0.0207	0.0019
	(0.0158)	(0.0102)	(0.0285)	(0.0201)	(0.0254)	(0.0083)
Missing TOEFL/IELTS	-0.0302*	0.0218	0.0544	0.0084	-0.1205***	0.0063
	(0.0168)	(0.0156)	(0.0517)	(0.0272)	(0.0434)	(0.0130)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic | DF | P-Value	37.4658	23	0.0290			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 5: Graduate School Lasso Logistic Regression Results

Depdendent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3330	-0.3123	-0.3498	-0.2378	
Winsorized Age	-0.0788	0.0266	0.0200		
Female	0.3352	0.5096	0.5068		-0.0654
U.S. African/Hispanic/Native	0.2638	0.8041	0.7157		
U.S. Asian	-0.2802	-0.2918		0.0413	0.1155
International Asian	-0.6744	-0.0726		0.0465	
International Other	0.4729	0.1946	0.7240	-0.1087	
GRE Verbal Percentile	0.0139	0.0252	0.0272		0.0343
GRE Quantitative Percentile	0.1349	0.0119	0.0187	0.0154	
GRE Analytic Percentile	0.0166	0.0208		0.0080	
Undergraduate GPA	1.9977	0.4621	0.7414	0.1714	
TOEFL/IELTS	0.4009	0.3295	0.5199	0.5151	
Elite U.S. University	0.6886	0.2709	0.3267		0.4606
Elite U.S. Liberal Arts College	0.8956	0.5199	0.5985		0.4470
Elite Foreign University	0.3724		0.0414	0.4343	
Other Foreign University					
Graduate Degree			0.0978		
Work Experience	0.2835	0.3447	0.3273		0.5559
#Professor Recommenders			0.3739	0.2322	
#Prolific Recommenders	0.2243	0.2362	0.1651	0.2213	0.0632
#Math Courses	0.1493				
Advanced Math	0.2272				
Missing GRE		-0.2304		-0.0527	-0.5416
Missing Undergraduate GPA					
Missing TOEFL/IELTS	-0.2364	0.2852	0.0796		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.2642	0.1456	0.1589	0.1486	0.1362
CV Pseudo-R^2	0.2297	0.0912	0.0592	0.0850	0.0537

Table 6: Simulated Wald Test Results

(a) Same Determinants

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3848***	-1.5822***	-28.5524***	-1.3677***	-10.4586***	-1.9105***	-10.2610***	-1.8514***
	(1.1315)	(0.0956)	(1.4963)	(0.0500)	(0.7236)	(0.1070)	(0.4815)	(0.0628)
GRE Quantitative Percentile	0.1509***	0.0146***	0.3245***	0.0155***	0.1058***	0.0193***	0.1154***	0.0208***
	(0.0131)	(0.0012)	(0.0179)	(0.0007)	(0.0088)	(0.0014)	(0.0070)	(0.0011)
Gender	0.4714***	0.0455***	0.7880***	0.0377***	-0.1384	-0.0253	-0.0497	-0.0090
	(0.1475)	(0.0140)	(0.1696)	(0.0081)	(0.1095)	(0.0200)	(0.0895)	(0.0161)
Demographic 1	-0.6287***	-0.0607***	-1.8488***	-0.0886***	-0.3486**	-0.0637**	-0.3079***	-0.0556***
	(0.2088)	(0.0200)	(0.2445)	(0.0115)	(0.1467)	(0.0267)	(0.1123)	(0.0202)
Demographic 2	0.1531	0.0148	0.3703*	0.0177*	0.1756	0.0321	0.0282	0.0051
	(0.2072)	(0.0200)	(0.1929)	(0.0091)	(0.1558)	(0.0284)	(0.1069)	(0.0193)
Advanced Math	0.2889*	0.0279*			0.3267***	0.0597***		
	(0.1612)	(0.0156)			(0.1112)	(0.0202)		
Committee Score	0.2241***	0.0216***	0.3994***	0.0191***	0.1026***	0.0187***	0.0757***	0.0137***
	(0.0413)	(0.0039)	(0.0506)	(0.0023)	(0.0287)	(0.0052)	(0.0228)	(0.0041)
Non-Economics 1			0.8376***	0.0401***			1.9483***	0.3515***
			(0.2297)	(0.0107)			(0.1236)	(0.0205)
Observations | Pseudo-R^2	5000	0.3251			5000	0.1431		
Parameters | LL | AIC	14	-1130.4	2288.8		14	-2691.4	5410.8	
Wald Statistic | DF | P-Value	3.1018	5	0.6843		3.0042	5	0.6993	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Different Determinants

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3291***	-1.5784***	-8.4753***	-0.3642***	-10.4581***	-1.9104***	-3.2236***	-0.5997***
	(1.1274)	(0.0955)	(0.8673)	(0.0412)	(0.7237)	(0.1070)	(0.4018)	(0.0728)
GRE Quantitative Percentile	0.1502***	0.0145***	0.0402***	0.0017***	0.1058***	0.0193***	0.0034	0.0006
	(0.0131)	(0.0012)	(0.0128)	(0.0006)	(0.0088)	(0.0014)	(0.0062)	(0.0012)
Gender	0.4702***	0.0454***	0.5942***	0.0255***	-0.1372	-0.0251	0.0375	0.0070
	(0.1473)	(0.0140)	(0.1820)	(0.0079)	(0.1095)	(0.0200)	(0.0868)	(0.0161)
Demographic 1	-0.6246***	-0.0604***	-0.2541	-0.0109	-0.3544**	-0.0647**	-0.1152	-0.0214
	(0.2085)	(0.0200)	(0.2393)	(0.0103)	(0.1467)	(0.0267)	(0.1103)	(0.0205)
Demographic 2	0.1555	0.0150	0.0081	0.0003	0.1713	0.0313	-0.0562	-0.0105
	(0.2070)	(0.0200)	(0.2109)	(0.0091)	(0.1557)	(0.0284)	(0.1030)	(0.0192)
Advanced Math	0.2886*	0.0279*			0.3284***	0.0600***		
	(0.1611)	(0.0156)			(0.1112)	(0.0202)		
Committee Score	0.2241***	0.0217***	0.3817***	0.0164***	0.1026***	0.0187***	0.1808***	0.0336***
	(0.0413)	(0.0039)	(0.0528)	(0.0025)	(0.0287)	(0.0052)	(0.0228)	(0.0041)
Non-Economics 1			-0.0452	-0.0019			1.1393***	0.2120***
			(0.1995)	(0.0086)			(0.1219)	(0.0219)
Observations | Pseudo-R^2	5000	0.1488			5000	0.0758		
Parameters | LL | AIC	14	-1155.7	2339.5		14	-2747.2	5522.4	
Wald Statistic | DF | P-Value	112.5897	5	0.0000		124.7379	5	0.0000	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

B Simulated Structural Results

In this appendix, I present structural estimation results from two simulated datasets, one in which the committee is nearly optimal and one in which it is not. The data-generating parameters for the means, variances, and correlations across the independent variables, as well as the parameters governing the effects of the independent variables on success, are meant to resemble those of an actual admissions dataset. In the (nearly) optimal dataset, the parameters governing the effect of the independent variables on acceptance are chosen to be similar to the success parameters. In contrast, the suboptimal committee discriminates against *Demographic 1*, undervalues *Advanced Math*, and overvalues *Committee Score*. The goal of this exercise is to verify that the estimated deviation gains are larger for the suboptimal dataset than the optimal dataset.

In Tables 7 and 8, I present the unrestricted and restricted dynamic estimates, respectively. Because I have two z variables, namely *Advanced Math* and *Committee Score*, I have two PDFs to estimate. Committee score is a continuous z, and since I parameterize continuous z's with the normal distribution, I have an additional σ_F parameter that equals the square root of the distribution's mean squared error. I also let *Committee Score* depend on *Advanced Math*, so I can estimate the joint distribution of the two z's without assuming the z's are independent. As usual, I do not assume independent z's. The simulated data are such that *GRE Quantitative Percentile*, *Advanced Math*, and *Program Rank* significantly predict success at the 1% level.[3]

The unrestricted and restricted static estimates are in Tables 9 and 10, respectively. Although the data are simulated, Table 10 contains an interesting result. There are two years in each of the simulated datasets: one in which the committee transitions 45–50% of the applicants to Round 2, and one in which it only transitions 25–30%. Essentially, in *Year Transition More*, the committee is more concerned with admitting a strong class than saving time and effort.[4] And in that year, as expected, the Round 1 cutoff probability is lower than the Round 2 cutoff probability. That is, it is easier to move on to Round 2 than it ultimately is to get into the program. However, in the other year, which is represented by the intercept term, the Round 1 cutoff is greater than the Round 2 cutoff, meaning the $P(\text{Success}|x,t,\overline{k})$ of the marginal applicant rejected in Round 1 is greater than the $P(\text{Success}|x,z,t,\overline{k})$ of the marginal applicant accepted into the program.

How can this be? Sometimes when a committee transitions a seemingly strong applicant to Round 2, the applicant's z reveals that they are weaker than previously thought. For example, the applicant may have a negative recommendation letter. If the committee transitions too few applicants to Round 2, it can end up in an unfortunate situation where it threw out so many files that it now has to accept those weaker applicants. As a simple example, suppose there are three applicants: A, B, and C. The committee has time to read two applicants' files and space to accept one applicant. Suppose that overall, A is the best applicant and C is the worst, but A

[3] As in the actual data, *Program Rank*'s sign is negative because the best rank is 1, and the worst rank is 150.

[4] In *Year Transition More*, the applicants also have higher matriculation probabilities on average, so this variable is called *Year Transition/Matriculate More* in the matriculation models in this appendix.

is the worst in objective variables. Then the committee will reject A in Round 1 and accept B, who is actually worse than A. In the simulated data, the minimum percentage of applications the committee must read in Round 2 to avoid this situation is between 25–30%, where the situation occurs, and 45–50%, where it does not. However, this situation does not occur in the actual data.

In Tables 11 and 12, I present estimates from the myopic model. Recall that the myopic model is a special case of the dynamic model in which the parameters for x in every block of $f_{z|x,t}$ are restricted to equal zero, while the intercepts, parameters for t, dependence parameters among the components of z, and variance parameters remain unrestricted. In this model, if an objective variable is associated with z in addition to being associated with success in its own right, the committee puts no extra weight on that variable in Round 1. Finally, the matriculation-dynamic estimates are in Tables 13 and 14. As with the dynamic estimates without matriculation, the simulated data are such that *GRE Quantitative Percentile*, *Advanced Math*, and *Program Rank* significantly predict success at the 1% level. But *Committee Score* significantly negatively predicts matriculation at the 1% level. The rationale is that applicants with higher committee scores likely have more outside offers and so are less likely to matriculate if accepted.

I conclude this appendix by presenting the deviation gains results on the simulated datasets. Per Figure 1, in the dataset where the committee is nearly optimal, the bootstrap-corrected models' CDFs for the acceptance set are within about one percentage point of the committee's CDF for that set. But in the suboptimal dataset, all the corrected models' CDFs strongly stochastically dominate the committee's CDF, and the corrected deviation gains are positive for all 500 perturbations. The reasons are that the committee is discriminating against *Demographic 1*, not putting enough weight on *Advanced Math*, and putting too much weight on *Committee Score*.

The results for the matriculation-dynamic and static models on the matriculation set are in Figure 2, and they are very similar to those in Figure 1. For the optimal dataset, the corrected gains are small and sometimes negative for individual perturbations, whereas for the suboptimal dataset, the corrected gains are large and always positive. In this figure, I also plot the CDF for the matriculation set of the dynamic model without matriculation. In the optimal dataset, *GRE Quantitative Percentile* negatively predicts matriculation, so the matriculation-dynamic committee selects relatively lower GREs in Round 1. This decision rule ends up not paying off in part because the simulation does not allow undershooting the target cohort size, a plausible simplification for a university like the one in this paper that is ranked highly enough to have never run out of applicants to accept in Round 2. However, in the actual data, even though the simulation does not allow undershooting, the matriculation-dynamic CDF is to the right of the dynamic CDF.

In Figures 3 and 4, I present the uncorrected deviation gains figures. Unlike in the actual data, the magnitude of the correction is only a small fraction of the uncorrected gains. The reason is that with 1,000 simulated observations and seven objective and subjective variables, the success probability parameter estimates are close to the true parameters. As such, the winner's curse is small. In contrast, in the actual data with 2,329 observations and 36 objective and subjective variables, there is more potential for overfitting, and the winner's curse is larger.

Table 7: Simulated Unrestricted Dynamic Estimates

(a) Optimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577**	-0.4303**	0.3695	-12.0690***	-2.2483***	-8.3663***	-1.6540***	-7.3172***	-1.0451***
	(0.7805)	(0.1891)	(1.4912)	(0.9758)	(0.1337)	(1.7459)	(0.3133)	(1.2885)	(0.1661)
GRE Quantitative Percentile	0.0251***	0.0061***	0.0510***	0.1260***	0.0235***	0.0911***	0.0180***	0.1028***	0.0147***
	(0.0094)	(0.0023)	(0.0170)	(0.0114)	(0.0017)	(0.0200)	(0.0036)	(0.0142)	(0.0017)
Gender	-0.2435*	-0.0596*	0.1807	0.0400	0.0074	0.0150	0.0030	0.2259	0.0323
	(0.1312)	(0.0319)	(0.2034)	(0.1525)	(0.0284)	(0.2415)	(0.0477)	(0.1762)	(0.0251)
Demographic 1	-0.0504	-0.0123	-0.6989**	0.0163	0.0030	-0.0503	-0.0100	-0.4384*	-0.0626*
	(0.1730)	(0.0424)	(0.2889)	(0.2031)	(0.0378)	(0.3722)	(0.0736)	(0.2358)	(0.0334)
Demographic 2	-0.3121	-0.0764	-0.4226	0.1678	0.0313	-0.2878	-0.0569	-0.4682*	-0.0669*
	(0.1930)	(0.0470)	(0.3254)	(0.2315)	(0.0431)	(0.4145)	(0.0818)	(0.2734)	(0.0387)
Advanced Math			1.1347***			0.2066	0.0408	0.7127***	0.1018***
			(0.1992)			(0.2496)	(0.0491)	(0.1791)	(0.0247)
Committee Score						0.0600	0.0119	-0.0790	-0.0113
						(0.0639)	(0.0126)	(0.0688)	(0.0098)
Missing Committee Score								0.0486	0.0069
								(0.1969)	(0.0281)
Year Transition More	-0.2523**	-0.0618**	0.1483	1.1441***	0.2131***	-1.1114***	-0.2197***	-0.3543**	-0.0506**
	(0.1282)	(0.0311)	(0.2025)	(0.1501)	(0.0250)	(0.2446)	(0.0435)	(0.1794)	(0.0254)
Program Rank								-0.0302***	-0.0043***
								(0.0018)	(0.0002)
$\sqrt{\text{MSE}}$			1.8393***						
			(0.0614)						
Obs | Param | LL | AIC	1000	38	-2602.7	5281.5					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577**	-0.4303**	1.2187	-10.3454***	-2.0151***	-4.8695***	-1.0694***	-7.3017***	-1.0448***
	(0.7805)	(0.1891)	(1.4016)	(0.9474)	(0.1440)	(1.5490)	(0.3252)	(1.2721)	(0.1635)
GRE Quantitative Percentile	0.0251***	0.0061***	0.0431***	0.1103***	0.0215***	0.0488***	0.0107***	0.1021***	0.0146***
	(0.0094)	(0.0023)	(0.0164)	(0.0111)	(0.0018)	(0.0180)	(0.0038)	(0.0140)	(0.0017)
Gender	-0.2435*	-0.0596*	0.1707	-0.0472	-0.0092	0.1746	0.0383	0.2087	0.0299
	(0.1312)	(0.0319)	(0.2037)	(0.1485)	(0.0289)	(0.2312)	(0.0506)	(0.1755)	(0.0251)
Demographic 1	-0.0504	-0.0123	-0.7422***	-0.4655**	-0.0907**	-0.5625*	-0.1235*	-0.4120*	-0.0590*
	(0.1730)	(0.0424)	(0.2732)	(0.1978)	(0.0382)	(0.3192)	(0.0689)	(0.2351)	(0.0334)
Demographic 2	-0.3121	-0.0764	-0.4868*	0.1277	0.0249	-0.1757	-0.0386	-0.3798	-0.0543
	(0.1930)	(0.0470)	(0.2925)	(0.2186)	(0.0426)	(0.3381)	(0.0742)	(0.2696)	(0.0384)
Advanced Math			1.1482***			-0.0098	-0.0021	0.7178***	0.1027***
			(0.1999)			(0.2366)	(0.0520)	(0.1795)	(0.0248)
Committee Score						0.1714***	0.0376***	-0.0769	-0.0110
						(0.0604)	(0.0128)	(0.0684)	(0.0098)
Missing Committee Score								0.1129	0.0162
								(0.1909)	(0.0273)
Year Transition More	-0.2523**	-0.0618**	0.0871	1.0879***	0.2119***	-0.7286***	-0.1600***	-0.3741**	-0.0535**
	(0.1282)	(0.0311)	(0.2014)	(0.1466)	(0.0255)	(0.2324)	(0.0485)	(0.1781)	(0.0253)
Program Rank								-0.0309***	-0.0044***
								(0.0019)	(0.0002)
$\sqrt{\text{MSE}}$			1.8362***						
			(0.0603)						
Obs | Param | LL | AIC	1000	38	-2648.4	5372.9					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 8: Simulated Restricted Dynamic Estimates

(a) Optimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-1.9394**	0.3771	-0.1457**	0.8644	0.2458	56.11%	-7.8112***	
	(0.7791)	(1.4942)	(0.0663)	(0.7591, 0.9843)	(0.2300)	(44.89%, 66.74%)	(1.1221)	
GRE Quantitative Percentile	0.0273***	0.0509***					0.1044***	
	(0.0094)	(0.0171)					(0.0128)	
Gender	-0.2478*	0.1808					0.0937	
	(0.1310)	(0.2034)					(0.1163)	
Demographic 1	-0.0432	-0.6993**					-0.2023	
	(0.1727)	(0.2904)					(0.1559)	
Demographic 2	-0.3026	-0.4231					-0.2735	
	(0.1930)	(0.3260)					(0.1817)	
Advanced Math		1.1347***					0.4597***	
		(0.1992)					(0.1452)	
Committee Score							-0.0021	
							(0.0436)	
Missing Committee Score							0.0240	
							(0.1891)	
Year Transition More	-0.2448*	0.1481	-0.1684***	0.7304	0.4096	65.82%	-0.3690**	
	(0.1281)	(0.2023)	(0.0637)	(0.6462, 0.8256)	(0.2744)	(57.07%, 73.62%)	(0.1763)	
Program Rank							-0.0300***	
							(0.0018)	
$\sqrt{\text{MSE}}$		1.8393***						
		(0.0614)						
Sigma								0.1647***
								(0.0242)
Obs | Param | LL | AIC	1000	29	-2642.8	5343.7				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-1.9242**	1.0267	-0.0633	0.9387	0.1263	53.15%	-7.7219***	
	(0.7773)	(1.4057)	(0.0892)	(0.7882, 1.1179)	(0.2508)	(40.97%, 64.97%)	(1.1783)	
GRE Quantitative Percentile	0.0271***	0.0453***					0.1006***	
	(0.0093)	(0.0165)					(0.0134)	
Gender	-0.2513*	0.1636					0.1366	
	(0.1310)	(0.2039)					(0.1297)	
Demographic 1	-0.0530	-0.7462***					-0.5186***	
	(0.1728)	(0.2730)					(0.1682)	
Demographic 2	-0.3004	-0.4758					-0.2123	
	(0.1928)	(0.2928)					(0.1992)	
Advanced Math		1.1476***					0.4443***	
		(0.1999)					(0.1595)	
Committee Score							0.0635	
							(0.0480)	
Missing Committee Score							0.1011	
							(0.1862)	
Year Transition More	-0.2466*	0.0910	-0.2559***	0.7268	0.3083	60.70%	-0.3596**	
	(0.1281)	(0.2010)	(0.0710)	(0.6177, 0.8551)	(0.3058)	(50.39%, 70.13%)	(0.1749)	
Program Rank							-0.0303***	
							(0.0018)	
$\sqrt{\text{MSE}}$		1.8362***						
		(0.0603)						
Sigma								0.2258***
								(0.0411)
Obs | Param | LL | AIC	1000	29	-2686.1	5430.2				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 9: Simulated Unrestricted Static Estimates

(a) Optimal Committee

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-12.0690***	-2.2483***	-8.3663***	-1.6540***	-7.3875***	-1.0821***	-7.3172***	-1.0451***
	(0.9758)	(0.1337)	(1.7459)	(0.3133)	(1.1197)	(0.1406)	(1.2885)	(0.1661)
GRE Quantitative Percentile	0.1260***	0.0235***	0.0911***	0.0180***	0.1043***	0.0153***	0.1028***	0.0147***
	(0.0114)	(0.0017)	(0.0200)	(0.0036)	(0.0134)	(0.0016)	(0.0142)	(0.0017)
Gender	0.0400	0.0074	0.0150	0.0030	0.1580	0.0231	0.2259	0.0323
	(0.1525)	(0.0284)	(0.2415)	(0.0477)	(0.1732)	(0.0254)	(0.1762)	(0.0251)
Demographic 1	0.0163	0.0030	-0.0503	-0.0100	-0.4556**	-0.0667**	-0.4384*	-0.0626*
	(0.2031)	(0.0378)	(0.3722)	(0.0736)	(0.2269)	(0.0328)	(0.2358)	(0.0334)
Demographic 2	0.1678	0.0313	-0.2878	-0.0569	-0.5153*	-0.0755**	-0.4682*	-0.0669*
	(0.2315)	(0.0431)	(0.4145)	(0.0818)	(0.2641)	(0.0383)	(0.2734)	(0.0387)
Advanced Math			0.2066	0.0408			0.7127***	0.1018***
			(0.2496)	(0.0491)			(0.1791)	(0.0247)
Committee Score			0.0600	0.0119			-0.0790	-0.0113
			(0.0639)	(0.0126)			(0.0688)	(0.0098)
Missing Committee Score							0.0486	0.0069
							(0.1969)	(0.0281)
Year Transition More	1.1441***	0.2131***	-1.1114***	-0.2197***	-0.3942**	-0.0577**	-0.3543**	-0.0506**
	(0.1501)	(0.0250)	(0.2446)	(0.0435)	(0.1689)	(0.0244)	(0.1794)	(0.0254)
Program Rank					-0.0299***	-0.0044***	-0.0302***	-0.0043***
					(0.0018)	(0.0002)	(0.0018)	(0.0002)
Obs | Param | LL | AIC	1000	31	-1632.5	3327.1				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-10.3454***	-2.0151***	-4.8695***	-1.0694***	-7.2362***	-1.0632***	-7.3017***	-1.0448***
	(0.9474)	(0.1440)	(1.5490)	(0.3252)	(1.1209)	(0.1418)	(1.2721)	(0.1635)
GRE Quantitative Percentile	0.1103***	0.0215***	0.0488***	0.0107***	0.1026***	0.0151***	0.1021***	0.0146***
	(0.0111)	(0.0018)	(0.0180)	(0.0038)	(0.0134)	(0.0016)	(0.0140)	(0.0017)
Gender	-0.0472	-0.0092	0.1746	0.0383	0.1408	0.0207	0.2087	0.0299
	(0.1485)	(0.0289)	(0.2312)	(0.0506)	(0.1727)	(0.0254)	(0.1755)	(0.0251)
Demographic 1	-0.4655**	-0.0907**	-0.5625*	-0.1235*	-0.4199*	-0.0617*	-0.4120*	-0.0590*
	(0.1978)	(0.0382)	(0.3192)	(0.0689)	(0.2285)	(0.0332)	(0.2351)	(0.0334)
Demographic 2	0.1277	0.0249	-0.1757	-0.0386	-0.4343*	-0.0638*	-0.3798	-0.0543
	(0.2186)	(0.0426)	(0.3381)	(0.0742)	(0.2605)	(0.0380)	(0.2696)	(0.0384)
Advanced Math			-0.0098	-0.0021			0.7178***	0.1027***
			(0.2366)	(0.0520)			(0.1795)	(0.0248)
Committee Score			0.1714***	0.0376***			-0.0769	-0.0110
			(0.0604)	(0.0128)			(0.0684)	(0.0098)
Missing Committee Score							0.1129	0.0162
							(0.1909)	(0.0273)
Year Transition More	1.0879***	0.2119***	-0.7286***	-0.1600***	-0.4350**	-0.0639***	-0.3741**	-0.0535**
	(0.1466)	(0.0255)	(0.2324)	(0.0485)	(0.1689)	(0.0244)	(0.1781)	(0.0253)
Program Rank					-0.0305***	-0.0045***	-0.0309***	-0.0044***
					(0.0019)	(0.0002)	(0.0019)	(0.0002)
Obs | Param | LL | AIC	1000	31	-1673.0	3408.1				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 10: Simulated Restricted Static Estimates

(a) Optimal Committee

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	1.2114***	77.05%	0.3765	59.30%	-7.7989***	-6.9427***	
	(0.2502)	(67.28%, 84.58%)	(0.2666)	(46.35%, 71.08%)	(1.0685)	(1.2241)	
GRE Quantitative Percentile					0.1065***	0.0957***	
					(0.0128)	(0.0139)	
Gender					0.0694	0.1374	
					(0.1077)	(0.1373)	
Demographic 1					-0.1638	-0.3053	
					(0.1410)	(0.1944)	
Demographic 2					-0.0994	-0.3969*	
					(0.1623)	(0.2189)	
Advanced Math						0.5127***	
						(0.1456)	
Committee Score						-0.0104	
						(0.0458)	
Missing Committee Score						-0.0145	
						(0.1924)	
Year Transition More	-1.3691***	46.07%	0.6127*	72.89%	-0.3712**	-0.3819**	
	(0.2738)	(40.01%, 52.24%)	(0.3291)	(61.87%, 81.67%)	(0.1649)	(0.1814)	
Program Rank					-0.0298***	-0.0300***	
					(0.0018)	(0.0018)	
Sigma							0.2021***
							(0.0252)
Obs | Param | LL | AIC	1000	22	-1639.7	3323.5			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	1.5071***	81.86%	0.2881	57.15%	-7.9222***	-6.4145***	
	(0.3814)	(68.13%, 90.50%)	(0.2858)	(43.24%, 70.02%)	(1.1110)	(1.2821)	
GRE Quantitative Percentile					0.1102***	0.0864***	
					(0.0132)	(0.0144)	
Gender					0.0340	0.2000	
					(0.1208)	(0.1433)	
Demographic 1					-0.4496***	-0.5033***	
					(0.1586)	(0.1917)	
Demographic 2					-0.1170	-0.3225	
					(0.1795)	(0.2194)	
Advanced Math						0.4844***	
						(0.1548)	
Committee Score						0.0524	
						(0.0491)	
Missing Committee Score						0.0419	
						(0.1918)	
Year Transition More	-1.6799***	45.69%	0.3660	65.79%	-0.4035**	-0.3931**	
	(0.3946)	(39.03%, 52.51%)	(0.3462)	(53.86%, 76.01%)	(0.1655)	(0.1786)	
Program Rank					-0.0303***	-0.0304***	
					(0.0019)	(0.0018)	
Sigma							0.2573***
							(0.0392)
Obs | Param | LL | AIC	1000	22	-1687.8	3419.7			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 11: Simulated Unrestricted Myopic Estimates

(a) Optimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.1765**	0.0439**	4.4154***	-12.0690***	-2.2483***	-8.3663***	-1.6540***	-7.3172***	-1.0451***
	(0.0898)	(0.0222)	(0.1909)	(0.9758)	(0.1337)	(1.7459)	(0.3133)	(1.2885)	(0.1661)
GRE Quantitative Percentile				0.1260***	0.0235***	0.0911***	0.0180***	0.1028***	0.0147***
				(0.0114)	(0.0017)	(0.0200)	(0.0036)	(0.0142)	(0.0017)
Gender				0.0400	0.0074	0.0150	0.0030	0.2259	0.0323
				(0.1525)	(0.0284)	(0.2415)	(0.0477)	(0.1762)	(0.0251)
Demographic 1				0.0163	0.0030	-0.0503	-0.0100	-0.4384*	-0.0626*
				(0.2031)	(0.0378)	(0.3722)	(0.0736)	(0.2358)	(0.0334)
Demographic 2				0.1678	0.0313	-0.2878	-0.0569	-0.4682*	-0.0669*
				(0.2315)	(0.0431)	(0.4145)	(0.0818)	(0.2734)	(0.0387)
Advanced Math			1.1642***			0.2066	0.0408	0.7127***	0.1018***
			(0.1981)			(0.2496)	(0.0491)	(0.1791)	(0.0247)
Committee Score						0.0600	0.0119	-0.0790	-0.0113
						(0.0639)	(0.0126)	(0.0688)	(0.0098)
Missing Committee Score								0.0486	0.0069
								(0.1969)	(0.0281)
Year Transition More	-0.2485**	-0.0618**	0.1348	1.1441***	0.2131***	-1.1114***	-0.2197***	-0.3543**	-0.0506**
	(0.1268)	(0.0313)	(0.2040)	(0.1501)	(0.0250)	(0.2446)	(0.0435)	(0.1794)	(0.0254)
Program Rank								-0.0302***	-0.0043***
								(0.0018)	(0.0002)
$\sqrt{\text{MSE}}$			1.8730***						
			(0.0618)						
Obs | Param | LL | AIC	1000	30	-2617.5	5295.0					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.1765**	0.0439**	4.5276***	-10.3454***	-2.0151***	-4.8695***	-1.0694***	-7.3017***	-1.0448***
	(0.0898)	(0.0222)	(0.1915)	(0.9474)	(0.1440)	(1.5490)	(0.3252)	(1.2721)	(0.1635)
GRE Quantitative Percentile				0.1103***	0.0215***	0.0488***	0.0107***	0.1021***	0.0146***
				(0.0111)	(0.0018)	(0.0180)	(0.0038)	(0.0140)	(0.0017)
Gender				-0.0472	-0.0092	0.1746	0.0383	0.2087	0.0299
				(0.1485)	(0.0289)	(0.2312)	(0.0506)	(0.1755)	(0.0251)
Demographic 1				-0.4655**	-0.0907**	-0.5625*	-0.1235*	-0.4120*	-0.0590*
				(0.1978)	(0.0382)	(0.3192)	(0.0689)	(0.2351)	(0.0334)
Demographic 2				0.1277	0.0249	-0.1757	-0.0386	-0.3798	-0.0543
				(0.2186)	(0.0426)	(0.3381)	(0.0742)	(0.2696)	(0.0384)
Advanced Math			1.1789***			-0.0098	-0.0021	0.7178***	0.1027***
			(0.1973)			(0.2366)	(0.0520)	(0.1795)	(0.0248)
Committee Score						0.1714***	0.0376***	-0.0769	-0.0110
						(0.0604)	(0.0128)	(0.0684)	(0.0098)
Missing Committee Score								0.1129	0.0162
								(0.1909)	(0.0273)
Year Transition More	-0.2485**	-0.0618**	0.0712	1.0879***	0.2119***	-0.7286***	-0.1600***	-0.3741**	-0.0535**
	(0.1268)	(0.0313)	(0.2044)	(0.1466)	(0.0255)	(0.2324)	(0.0485)	(0.1781)	(0.0253)
Program Rank								-0.0309***	-0.0044***
								(0.0019)	(0.0002)
$\sqrt{\text{MSE}}$			1.8655***						
			(0.0600)						
Obs | Param | LL | AIC	1000	30	-2662.4	5384.8					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 12: Simulated Restricted Myopic Estimates

(a) Optimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.1822**	4.4131***	-0.1459**	0.8642	0.2450	56.09%	-7.8232***	
	(0.0898)	(0.1915)	(0.0669)	(0.7581, 0.9852)	(0.2303)	(44.86%, 66.74%)	(1.1263)	
GRE Quantitative Percentile							0.1055***	
							(0.0128)	
Gender							0.0872	
							(0.1160)	
Demographic 1							-0.2089	
							(0.1545)	
Demographic 2							-0.2850	
							(0.1804)	
Advanced Math		1.1642***					0.4385***	
		(0.1982)					(0.1452)	
Committee Score							-0.0125	
							(0.0436)	
Missing Committee Score							0.0243	
							(0.1889)	
Year Transition More	-0.2418*	0.1344	-0.1671***	0.7312	0.4173	65.98%	-0.3689**	
	(0.1267)	(0.2039)	(0.0640)	(0.6463, 0.8273)	(0.2757)	(57.16%, 73.81%)	(0.1764)	
Program Rank							-0.0300***	
							(0.0018)	
$\sqrt{\text{MSE}}$		1.8730***						
		(0.0618)						
Sigma								0.1652***
								(0.0244)
Obs | Param | LL | AIC	1000	21	-2658.4	5358.7				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.1801**	4.5328***	-0.0597	0.9420	0.1255	53.13%	-7.7935***	
	(0.0898)	(0.1919)	(0.0912)	(0.7878, 1.1264)	(0.2525)	(40.87%, 65.03%)	(1.1728)	
GRE Quantitative Percentile							0.1025***	
							(0.0132)	
Gender							0.1327	
							(0.1299)	
Demographic 1							-0.5335***	
							(0.1674)	
Demographic 2							-0.2291	
							(0.1982)	
Advanced Math		1.1784***					0.4297***	
		(0.1973)					(0.1610)	
Committee Score							0.0501	
							(0.0488)	
Missing Committee Score							0.1041	
							(0.1858)	
Year Transition More	-0.2436*	0.0726	-0.2565***	0.7289	0.3183	60.92%	-0.3616**	
	(0.1267)	(0.2043)	(0.0714)	(0.6175, 0.8605)	(0.3098)	(50.44%, 70.47%)	(0.1752)	
Program Rank							-0.0303***	
							(0.0018)	
$\sqrt{\text{MSE}}$		1.8654***						
		(0.0599)						
Sigma								0.2283***
								(0.0424)
Obs | Param | LL | AIC	1000	21	-2701.0	5444.0				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 13: Simulated Unrestricted Matriculation-Dynamic Estimates

(a) Optimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577**	-0.4303**	0.3695	-12.0690***	-2.2483***	-8.3663***	-1.6540***	-7.3172***	-1.0451***	10.8487*	1.1240**
	(0.7805)	(0.1891)	(1.4912)	(0.9758)	(0.1337)	(1.7459)	(0.3133)	(1.2885)	(0.1661)	(5.7339)	(0.5653)
GRE Quantitative Percentile	0.0251***	0.0061***	0.0510***	0.1260***	0.0235***	0.0911***	0.0180***	0.1028***	0.0147***	-0.1049*	-0.0109*
	(0.0094)	(0.0023)	(0.0170)	(0.0114)	(0.0017)	(0.0200)	(0.0036)	(0.0142)	(0.0017)	(0.0630)	(0.0063)
Gender	-0.2435*	-0.0596*	0.1807	0.0400	0.0074	0.0150	0.0030	0.2259	0.0323	-0.9225	-0.0956*
	(0.1312)	(0.0319)	(0.2034)	(0.1525)	(0.0284)	(0.2415)	(0.0477)	(0.1762)	(0.0251)	(0.5834)	(0.0575)
Demographic 1	-0.0504	-0.0123	-0.6989**	0.0163	0.0030	-0.0503	-0.0100	-0.4384*	-0.0626*	1.2693	0.1315
	(0.1730)	(0.0424)	(0.2889)	(0.2031)	(0.0378)	(0.3722)	(0.0736)	(0.2358)	(0.0334)	(0.8109)	(0.0828)
Demographic 2	-0.3121	-0.0764	-0.4226	0.1678	0.0313	-0.2878	-0.0569	-0.4682*	-0.0669*	0.2958	0.0306
	(0.1930)	(0.0470)	(0.3254)	(0.2315)	(0.0431)	(0.4145)	(0.0818)	(0.2734)	(0.0387)	(0.9734)	(0.1014)
Advanced Math			1.1347***			0.2066	0.0408	0.7127***	0.1018***	0.3819	0.0396
			(0.1992)			(0.2496)	(0.0491)	(0.1791)	(0.0247)	(0.5414)	(0.0557)
Committee Score						0.0600	0.0119	-0.0790	-0.0113	-1.0758***	-0.1115***
						(0.0639)	(0.0126)	(0.0688)	(0.0098)	(0.2376)	(0.0153)
Missing Committee Score								0.0486	0.0069		
								(0.1969)	(0.0281)		
Year Transition/Matriculate More	-0.2523**	-0.0618**	0.1483	1.1441***	0.2131***	-1.1114***	-0.2197***	-0.3543**	-0.0506**	5.2621***	0.5452***
	(0.1282)	(0.0311)	(0.2025)	(0.1501)	(0.0250)	(0.2446)	(0.0435)	(0.1794)	(0.0254)	(0.8695)	(0.0260)
Program Rank								-0.0302***	-0.0043***		
								(0.0018)	(0.0002)		
$\sqrt{\text{MSE}}$			1.8393***								
			(0.0614)								
Obs | Param | LL | AIC	1000	46	-2644.2	5380.5							

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577**	-0.4303**	1.2187	-10.3454***	-2.0151***	-4.8695***	-1.0694***	-7.3017***	-1.0448***	5.7452	0.5714
	(0.7805)	(0.1891)	(1.4016)	(0.9474)	(0.1440)	(1.5490)	(0.3252)	(1.2721)	(0.1635)	(4.2604)	(0.4164)
GRE Quantitative Percentile	0.0251***	0.0061***	0.0431***	0.1103***	0.0215***	0.0488***	0.0107***	0.1021***	0.0146***	-0.0443	-0.0044
	(0.0094)	(0.0023)	(0.0164)	(0.0111)	(0.0018)	(0.0180)	(0.0038)	(0.0140)	(0.0017)	(0.0468)	(0.0046)
Gender	-0.2435*	-0.0596*	0.1707	-0.0472	-0.0092	0.1746	0.0383	0.2087	0.0299	-1.0062*	-0.1001*
	(0.1312)	(0.0319)	(0.2037)	(0.1485)	(0.0289)	(0.2312)	(0.0506)	(0.1755)	(0.0251)	(0.5693)	(0.0535)
Demographic 1	-0.0504	-0.0123	-0.7422***	-0.4655**	-0.0907**	-0.5625*	-0.1235*	-0.4120*	-0.0590*	0.8111	0.0807
	(0.1730)	(0.0424)	(0.2732)	(0.1978)	(0.0382)	(0.3192)	(0.0689)	(0.2351)	(0.0334)	(0.7062)	(0.0712)
Demographic 2	-0.3121	-0.0764	-0.4868*	0.1277	0.0249	-0.1757	-0.0386	-0.3798	-0.0543	0.7188	0.0715
	(0.1930)	(0.0470)	(0.2925)	(0.2186)	(0.0426)	(0.3381)	(0.0742)	(0.2696)	(0.0384)	(0.7190)	(0.0724)
Advanced Math			1.1482***			-0.0098	-0.0021	0.7178***	0.1027***	-0.7520	-0.0748
			(0.1999)			(0.2366)	(0.0520)	(0.1795)	(0.0248)	(0.5452)	(0.0534)
Committee Score						0.1714***	0.0376***	-0.0769	-0.0110	-1.0032***	-0.0998***
						(0.0604)	(0.0128)	(0.0684)	(0.0098)	(0.2212)	(0.0150)
Missing Committee Score								0.1129	0.0162		
								(0.1909)	(0.0273)		
Year Transition/Matriculate More	-0.2523**	-0.0618**	0.0871	1.0879***	0.2119***	-0.7286***	-0.1600***	-0.3741**	-0.0535**	5.6048***	0.5574***
	(0.1282)	(0.0311)	(0.2014)	(0.1466)	(0.0255)	(0.2324)	(0.0485)	(0.1781)	(0.0253)	(0.9335)	(0.0301)
Program Rank								-0.0309***	-0.0044***		
								(0.0019)	(0.0002)		
$\sqrt{\text{MSE}}$			1.8362***								
			(0.0603)								
Obs | Param | LL | AIC	1000	46	-2693.0	5478.1							

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 14: Simulated Restricted Matriculation-Dynamic Estimates

(a) Optimal Committee

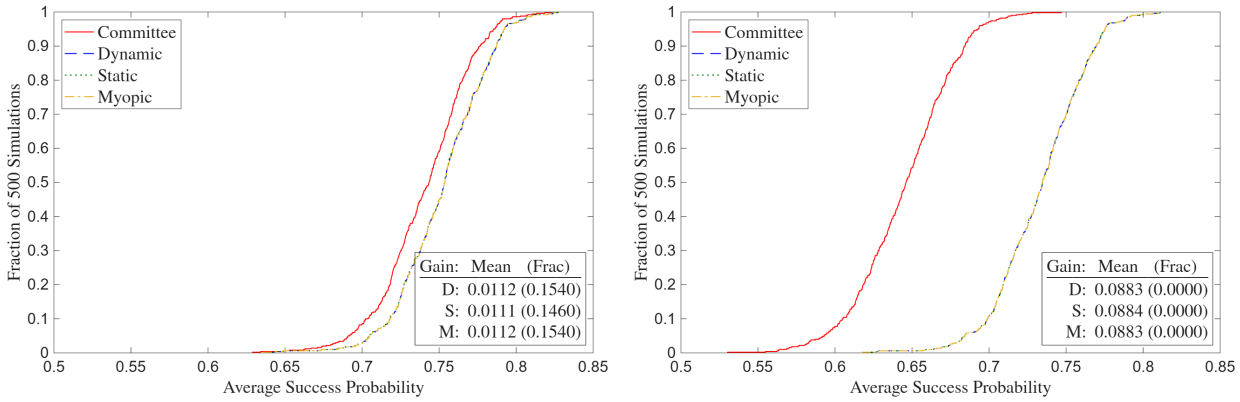
	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-1.8113**	1.2405	-1.4922***	0.2249	0.3506	58.68%	-7.7566***	2.9160	
	(0.7828)	(1.5031)	(0.1913)	(0.1546, 0.3272)	(0.2619)	(45.94%, 70.35%)	(1.1350)	(5.1107)	
GRE Quantitative Percentile	0.0257***	0.0408**					0.1057***	-0.0133	
	(0.0094)	(0.0173)					(0.0129)	(0.0562)	
Gender	-0.2464*	0.1731					0.1159	-0.9424*	
	(0.1312)	(0.2034)					(0.1353)	(0.5543)	
Demographic 1	-0.0487	-0.7009**					-0.2877	1.2129	
	(0.1729)	(0.2890)					(0.1882)	(0.8220)	
Demographic 2	-0.3098	-0.4291					-0.3732*	0.5064	
	(0.1929)	(0.3258)					(0.2125)	(0.9746)	
Advanced Math		1.1261***					0.4974***	0.3171	
		(0.2002)					(0.1521)	(0.5299)	
Committee Score							-0.0238	-1.0550***	
							(0.0447)	(0.2347)	
Missing Committee Score							0.0211		
							(0.1952)		
Year Transition/Matriculate More	-0.2441*	0.1538	-1.2324***	0.0656	0.4094	68.14%	-0.3789**	4.9595***	
	(0.1282)	(0.2051)	(0.2881)	(0.0355, 0.1212)	(0.3147)	(57.83%, 76.93%)	(0.1823)	(0.9084)	
Program Rank							-0.0301***		
							(0.0018)		
$\sqrt{\text{MSE}}$		1.8440***							
		(0.0615)							
Sigma									0.1872***
									(0.0329)
Obs | Param | LL | AIC	1000	37	-2707.6	5489.2					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

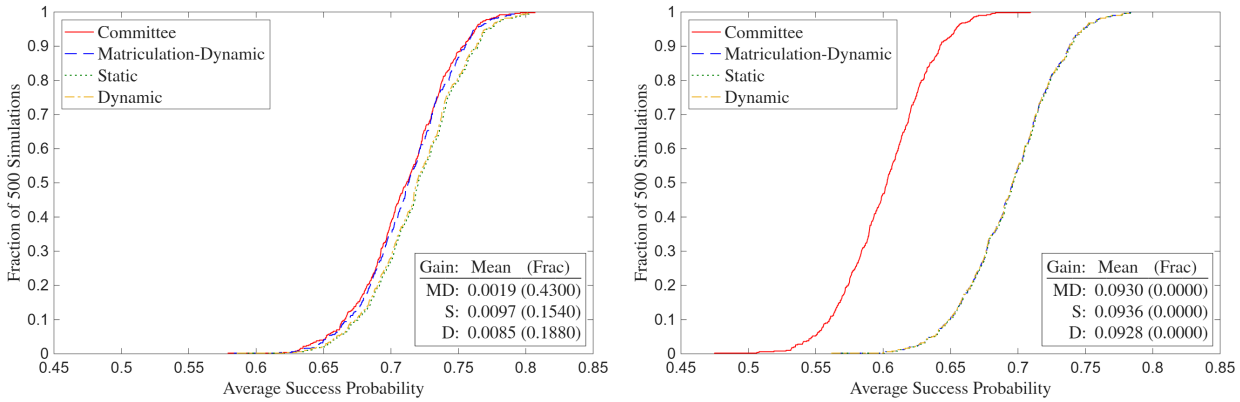
	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-1.7155**	1.8306	-1.0818***	0.3390	0.2612	56.49%	-7.7219***	0.2437	
	(0.7813)	(1.4228)	(0.2600)	(0.2036, 0.5643)	(0.3040)	(41.71%, 70.20%)	(1.1701)	(4.4569)	
GRE Quantitative Percentile	0.0246***	0.0357**					0.1020***	0.0200	
	(0.0094)	(0.0167)					(0.0133)	(0.0495)	
Gender	-0.2456*	0.1683					0.1877	-1.0474*	
	(0.1312)	(0.2035)					(0.1499)	(0.5402)	
Demographic 1	-0.0469	-0.7099***					-0.5368***	0.5144	
	(0.1731)	(0.2735)					(0.1983)	(0.7157)	
Demographic 2	-0.3097	-0.4873*					-0.3087	0.7381	
	(0.1929)	(0.2925)					(0.2284)	(0.7585)	
Advanced Math		1.1443***					0.5177***	-0.8396	
		(0.2001)					(0.1729)	(0.5352)	
Committee Score							0.0412	-0.9473***	
							(0.0545)	(0.2114)	
Missing Committee Score							0.0852		
							(0.1912)		
Year Transition/Matriculate More	-0.2478*	0.0862	-1.0168***	0.1226	0.3707	65.29%	-0.3780**	5.3618***	
	(0.1281)	(0.2022)	(0.2514)	(0.0646, 0.2327)	(0.4109)	(50.97%, 77.29%)	(0.1794)	(0.9378)	
Program Rank							-0.0305***		
							(0.0018)		
$\sqrt{\text{MSE}}$		1.8366***							
		(0.0602)							
Sigma									0.2943***
									(0.0717)
Obs | Param | LL | AIC	1000	37	-2747.6	5569.2					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.



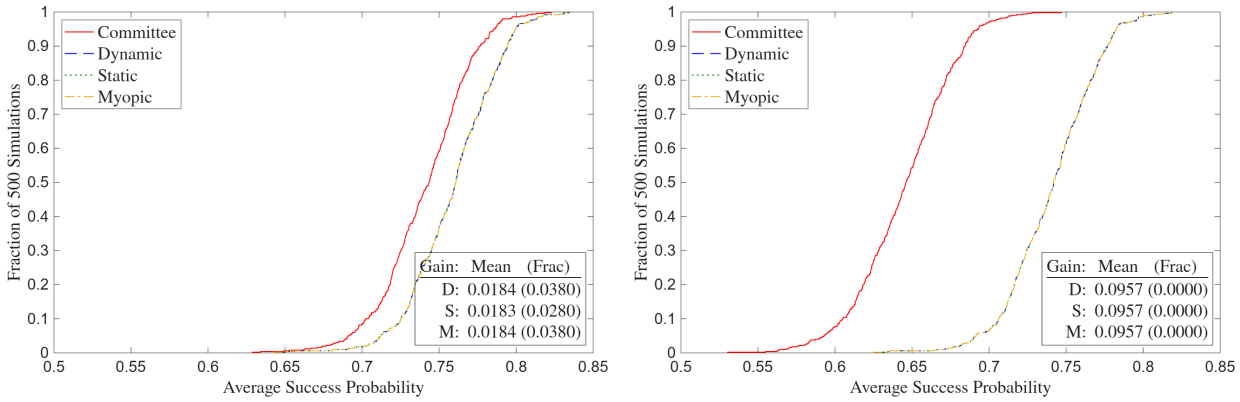
(a) Optimal Committee (b) Suboptimal Committee

Figure 1: Simulated Deviation Gains (Acceptance Set)



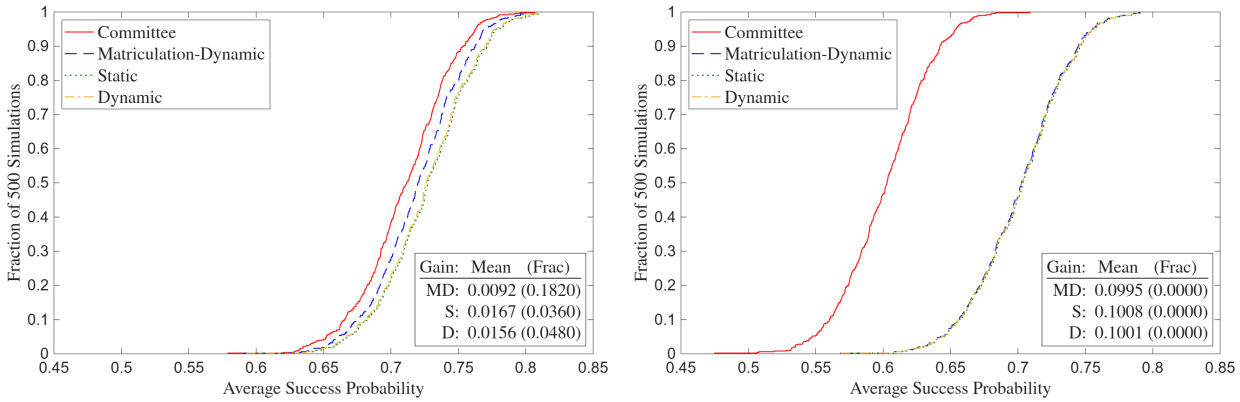
(a) Optimal Committee (b) Suboptimal Committee

Figure 2: Simulated Deviation Gains (Matriculation Set)



(a) Optimal Committee (b) Suboptimal Committee

Figure 3: Uncorrected Simulated Deviation Gains (Acceptance Set)



(a) Optimal Committee (b) Suboptimal Committee

Figure 4: Uncorrected Simulated Deviation Gains (Matriculation Set)

C Waitlist Models

In the matriculation-dynamic model in Section 4.2.1, the committee considers matriculation probabilities in Round 1. However, the degree to which it does so is small enough that the static model without matriculation serves as a static approximation for the matriculation-dynamic model. Committee members at the university featured in this paper also indicate that the committee does not consider matriculation probabilities in Round 1. The reason is that this university is ranked highly enough that it can fill its cohort with applicants who have strong objective and subjective variables, leaving Round 2 with enough slack to be selective on success regardless of Round 1's matriculation weighting. However, this university does consider matriculation probabilities after Round 1, where additional variables, such as *Letters Terms (%) PC1*, predict matriculation.[5]

Therefore, in this appendix, I introduce the waitlist and expand the cutoff rule framework from two rounds to three. Success probabilities equal, the committee favors likely matriculants in Round 2 to avoid losing high-quality applicants who would have matriculated by suboptimally keeping those applicants waiting until Round 3 for an acceptance. I also make the simplifying assumption that matriculation probabilities do not enter the Round 1 objective.

C.1 Asymmetric Waitlist Models

An admissions committee does not send out all its acceptances at once, but rather in batches. It starts by accepting its most preferred applicants, and then it puts the others who survived Round 1 on a waitlist. Then provided not all its early acceptances matriculate, it accepts applicants off the waitlist and ultimately moves down the waitlist until it has filled its cohort. I model this process as concisely as possible by increasing the number of rounds to three. As shown in Figure 5, the committee observes only the objective variables in Round 1 and either transitions the applicant to Round 2 or rejects them out of courtesy. But in Round 2, instead of accepting or rejecting the survivors based on the objective and subjective variables, it accepts (A) or waitlists (W) them.

Then in Round 3, it accepts or rejects the waitlisted applicants. Between Rounds 2 and 3, it observes how many of the early acceptances matriculate and makes its Round 3 decisions accordingly. If more early acceptances matriculate, it accepts fewer applicants off the waitlist because there are fewer openings left to fill. Intuitively, this situation is better than the other way around. More early acceptances matriculating means the committee gets more of its top choices. In contrast, if the committee has to accept many applicants off the waitlist, it may have to lower its standards because applicants it could have gotten with early offers may have already committed to other schools. Therefore, it should err on the side of giving Round 2 offers to applicants who are relatively likely to matriculate. That said, it would be incorrect to have the committee value

[5]Per Table 2 of Section 3, the matriculation-predicting variables in x are *Winsorized Age* (-), *U.S.* (+), *International Asian* (+), *GRE Quantitative Percentile* (-), and *Work Experience* (+). The matriculation-predicting variables in z are *Advanced Math* (-), *Letters Terms (%) PC1* (+), *CV Coding* (-), and *Essay Economics Terms (%)* (+).

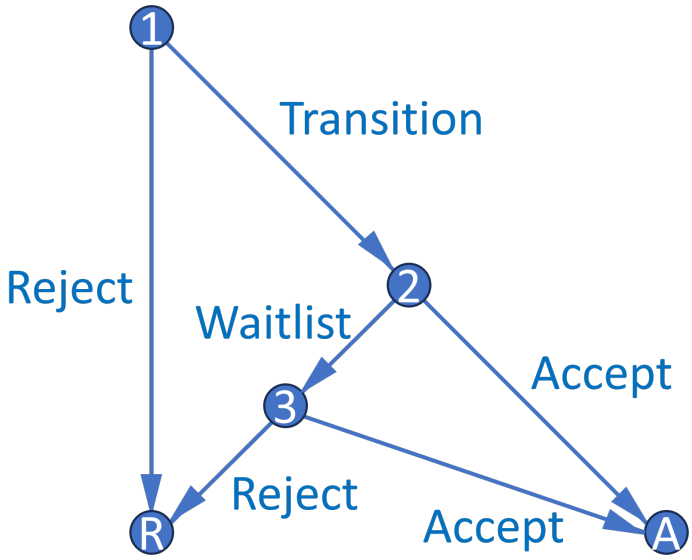


Figure 5: Waitlist Decision Graph

applicants by a combination of their success and matriculation probabilities, as that would be the cutoff rule solution to a portfolio choice problem with a constraint on the number of acceptances, not matriculants. The matriculation probabilities must instead appear elsewhere in the objective.

As such, I set up the model by letting the Round 3 cutoff probability be $P(y_3^* = S|\overline{m_{2A}})$ instead of $P(y_3^* = S|t)$. Here, $\overline{m_{2A}}$ is the average early accepted applicant's matriculation probability in a given year, which measures how likely the applicant is to matriculate if accepted. $P(\text{Cutoff})_3$ increases in $\overline{m_{2A}}$ because a larger $\overline{m_{2A}}$ means there will be fewer openings for waitlisted applicants. Given that, what information available at the start of Round 2 determines $\overline{m_{2A}}$? For all applicants in each year, I define $\ddot{m}_2$ as $\overline{m_2}$ but with m_2 excluded, where m_2 is the applicant's own matriculation probability. Within each year, a low m_2 necessarily implies a high $\ddot{m}_2$, since $\overline{m_2}$ is a within-year average. I then estimate $f_{\overline{m_{2A}}|\ddot{m}_2}$ on all transitioned applicants; the parameter estimate should be positive provided there exists variation in $\overline{m_2}$ across years. For example, by decreasing acceptance probabilities, Covid may have made applicants more likely to matriculate if accepted. I assume that the effect of the average matriculation probability of the applicants the committee can accept ($\ddot{m}_2$) on that of the applicants it accepts ($\overline{m_{2A}}$) is stationary across years, and therefore can be used to forecast changes in $\overline{m_{2A}}$ based on changes in $\ddot{m}_2$ resulting from changes in d_2.

The model implies that the optimal dynamic decision in Round 2 may not be the same as the optimal myopic one. Specifically, a committee may waitlist, as opposed to accept, an applicant with a low m_2 even if their success probability is greater than an applicant with a higher m_2. The reason is that compared with a high m_2 applicant, a low m_2 applicant has a relatively high $\ddot{m}_2$. That implies a high $\overline{F}_{\overline{m_{2A}}|\ddot{m}_2}$, which in turn implies a high Round 3 cutoff probability, Round 3 value function, and payoff for waitlisting the low m_2 applicant in Round 2. Suppose that a hypothetical

committee does not think dynamically; that is, removing these memory parameters decreases the AIC. However, if $\ddot{m}_2$ affects $\overline{F}_{\overline{m_{2A}}|\ddot{m}_2}$ and in turn $\mathbb{E}[P(\text{Cutoff})_3]$, perhaps the committee should.

That said, why is the committee not able to accept the low m_2 applicant in Round 2 and then, if this applicant does not matriculate, fill the additional opening with the high m_2 applicant that it necessarily waitlisted? There are two reasons why such a strategy may not pay off. First, all else equal, matriculation probabilities are expected to decline between Rounds 2 and 3 because applicants are more likely to go where they feel wanted; whether and by how much they decline is an empirical matter. If matriculation probabilities were to decline uniformly for likely vs. unlikely matriculants, the committee could still get away with accepting unlikely matriculants and filling any extra openings with likely matriculants. However, if applicants with high matriculation probabilities have more room for those matriculation probabilities to fall when the committee keeps them waiting until Round 3, then the better decision is to accept them in Round 2.

Second, matriculation probabilities can change unpredictably between Rounds 2 and 3 because applicants receive acceptances, waitlists, and rejections from multiple universities, communicate with faculty and graduate students at universities where they were accepted or waitlisted, and attend open houses. These events, which are unknown to the committee at the start of Round 2, make it so that waitlisting a likely matriculant under the belief that they will still matriculate if waitlisted is a risky endeavor; by Round 3, the committee may find out too late that this applicant is now unlikely to matriculate. Therefore, it is safer to accept this applicant in Round 2.[6]

As such, it is clear that accepting applicants who want to matriculate benefits the program. I also discount the expected Round 3 value with $\beta \in (0,1)$ so that the payoff from accepting an applicant in Round 3 is less than that from accepting them in Round 2. The discount factor captures the committee's perceived cost of deferring the final decision to Round 3 and can be interpreted as the chance of losing a representative applicant to another university; loosely, it reflects the ratio of expected matriculation probabilities for Round 3 vs. Round 2 acceptances. Of course, without a capacity constraint, the committee could give early acceptances to the low and high m_2 applicants, but because of the constraint, it must avoid overshooting its target cohort size. I now present the value functions for the waitlist-dynamic model, starting with Round 3:

$$v_3(x,z,t,\bar{k},\overline{m_{2A}},\varepsilon_3) = \max_{d_3 \in \{A,R\}} u_3(x,z,t,\bar{k},\overline{m_{2A}},d_3) + \varepsilon_3(d_3), \text{ where:} \tag{3}$$

$$u_3(x,z,t,\bar{k},\overline{m_{2A}},d_3) = P(y=S|x,z,t,\bar{k})\mathbb{1}(d_3=A) + P(y_3^*=S|\overline{m_{2A}})\mathbb{1}(d_3=R).$$

The Round 2 value function is:

$$v_2(x,z,t,\bar{k},\ddot{m}_2,\varepsilon_2) = \max_{d_2 \in \{A,W\}} u_2(x,z,t,\bar{k},\ddot{m}_2,d_2) + \varepsilon_2(d_2), \text{ where:} \tag{4}$$

[6] In Appendix C.3, I present a more general version of the model in which the Round 3 cutoff probability is a function of $\overline{m_{2A}}$ and $\overline{m_3}$, which is the average matriculation probability of the waitlisted applicants and depends on m_3, the *ex-ante* probability of matriculating if accepted in Round 3. In this model, the effect of matriculation probabilities on Round 2 acceptance probabilities is theoretically ambiguous and depends on the parameter values.

$$u_2(x,z,t,\bar{k},\ddot{m}_2,d_2) = P(y=S|x,z,t,\bar{k})\mathbb{1}(d_2=A) + \beta\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m_{2A}},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2]\mathbb{1}(d_2=W), \text{ where:}$$

$$\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m_{2A}},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2] = \int_{\widetilde{\overline{m_{2A}}}} \sigma \log\left(\sum_{\widetilde{d}_3\in\{A,R\}} e^{u_3(x,z,t,\bar{k},\widetilde{\overline{m_{2A}}},\widetilde{d}_3)/\sigma}\right) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2).$$

The Round 1 value function depends on whether I include a conditional expectation, as in the dynamic model, or whether I do not, as in the static model. I refer to the three-round model without a Round 1 conditional expectation as the waitlist-hybrid model, since it has a static Round 1 and dynamic Round 2. The waitlist-dynamic model's Round 1 value function is:

$$v_1(x,t,\bar{k},\varepsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,\bar{k},d_1) + \varepsilon_1(d_1), \text{ where:}$$

$$u_1(x,t,\bar{k},d_1) = \mathbb{E}[v_2(x,z,t,\bar{k},\ddot{m}_2,\varepsilon_2)|x,t,\bar{k}]\mathbb{1}(d_1=T) + L(y_1^*=S|t)\mathbb{1}(d_1=R), \text{ where:} \tag{5}$$

$$\mathbb{E}[v_2(x,z,t,\bar{k},\ddot{m}_2,\varepsilon_2)|x,t,\bar{k}] = \int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2\in\{A,W\}} e^{u_2(x,\widetilde{z},t,\bar{k},\ddot{m}_2(x,\widetilde{z},t),\widetilde{d}_2)/\sigma}\right) dF_{z|x,t}(\widetilde{z}|x,t).$$

In contrast, the utility function in the waitlist-hybrid's Round 1 value function is:

$$u_1(x,t,\bar{k},d_1) = P(y=S|x,t,\bar{k})\mathbb{1}(d_1=T) + P(y_1^*=S|t)\mathbb{1}(d_1=R). \tag{6}$$

Expressed as simply as possible, the restricted CCPs are:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states},d_r)/\sigma)}{\sum_{\widetilde{d}_r\in D_r}\exp(u_r(\text{states},\widetilde{d}_r)/\sigma)}. \tag{7}$$

The waitlist-dynamic model's unrestricted CCPs, success probabilities, and cutoff probabilities are $P_{d_1} = P\left(d_1 = T\middle|x,t;\theta_T\right)$, $P_{d_2} = P\left(d_2 = A\middle|x,z,\ddot{m}_2;\theta_{A_2}\right)$, $P_{d_3} = P\left(d_3 = A\middle|x,z,\overline{m_{2A}};\theta_{A_3}\right)$, $P_y = P\left(y = S\middle|x,z,t,\bar{k};\theta_S\right)$, $L_{y_1^*} = L\left(y_1^* = S\middle|t;\theta_{S_1^*}\right)$, and $P_{y_3^*} = P\left(y_3^* = S\middle|\overline{m_{2A}};\theta_{S_3^*}\right)$. These probabilities have the same specifications as in the dynamic model (see Equations (11)–(13) for the functional forms).[7] Let applicants 1 to N_W be waitlisted, N_W+1 to N_T accepted in Round 2, and N_T+1 to N rejected in Round 1. Then the unrestricted waitlist-dynamic likelihood is:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} f\left(z_n^c|z_n^{\neg c},x_n,t_n;\theta_{F_z}\right) f\left(\overline{m_{2A,n}}|\ddot{m}_{2,n};\theta_{F_m}\right)$$
$$P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_{A_2})^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_{A_2}))^{\mathbb{1}(d_{2,n}=W)} \prod_{n=1}^{N_W} P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}(1-P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)}$$
$$\prod_{n=N_T+1}^{N}\left(1-P_{d_{1,n}}(\theta_T)\right)\prod_{n=1}^{N} f\left(z_n^{\neg c}|x_n,t_n;\theta_{F_z}\right) P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}. \tag{8}$$

[7] $L_{y_1^*}$ is parameterized as exponential because it can exceed 1. However, in the waitlist-hybrid model, it is denoted as $P_{y_1^*}$ and parameterized as logistic. The waitlist-hybrid model also includes $P_{y_1}\left(y = S\middle|x,t,\bar{k};\theta_{S_1}\right)$.

I also estimate the parameters of $m_2 = P_{m_2}(\theta_{M_2})$, and therefore $\ddot{m}_2$, in a first-stage likelihood where I restrict the sample to Round 2 acceptances. The dependent variable equals 1 if the applicant matriculates and 0 otherwise, and the independent variables are x, z, and t. Because $\ddot{m}_2$ is a generated regressor in both the unrestricted and restricted likelihoods, I estimate standard errors per Murphy and Topel (1985) and Hardin (2002). The first-stage likelihood is:

$$\mathcal{L}^{M_2}_{N_T-N_W}(\theta_{M_2}) = \prod_{n=N_W+1}^{N_T} P_{m_{2,n}}(\theta_{M_2})^{\mathbb{1}(m_{2,n}=M)}(1-P_{m_{2,n}}(\theta_{M_2}))^{\mathbb{1}(m_{2,n}=D)}. \tag{9}$$

To show that the waitlist models' dynamics work as intended, I prove the following theorem:

Theorem 1. *Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. Suppose there are two arbitrary applicants in year t indexed by H and L, such that $P(y = S|x^H, z^H, t, \bar{k}) = P(y = S|x^L, z^L, t, \bar{k})$, but $m_2^H > m_2^L$. That is, the applicants have equal success probabilities, but H has the higher matriculation probability. Since $\ddot{m}_2$ is the average matriculation probability of all applicants in t with m_2 excluded, $\ddot{m}_2^H < \ddot{m}_2^L$ by construction. Given this inequality, I will show $P(d_2 = A|x^H, z^H, t, \bar{k}, \ddot{m}_2^H) > P(d_2 = A|x^L, z^L, t, \bar{k}, \ddot{m}_2^L)$. It is the case that:

$$\begin{aligned} P(d_2 = A|x,z,t,\bar{k},\ddot{m}_2) &= \frac{\exp(P(y=S|x,z,t,\bar{k})/\sigma)}{\exp(P(y=S|x,z,t,\bar{k})/\sigma) + \exp(\beta\mathbb{E}[v_3(\cdot)|\cdot]/\sigma)}, \text{ where} \\ \mathbb{E}[v_3(\cdot)|\cdot] &= \int_{\widetilde{\overline{m}_{2A}}} \sigma \log\left(e^{P(y=S|x,z,t,\bar{k})/\sigma} + e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right) dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2). \end{aligned} \tag{10}$$

The only place $\ddot{m}_2$ appears in $P(d_2 = A|x,z,t,\bar{k},\ddot{m}_2)$ is in the distribution of $\overline{m}_{2A}$ conditional on $\ddot{m}_2$. I parameterize $\overline{m}_{2A}|\ddot{m}_2 \sim \mathcal{N}(\ddot{m}_2\beta_{\ddot{m}_2}, \sigma^2_{\ddot{m}_2})$, where by assumption $\beta_{\ddot{m}_2} > 0$. $\beta_{\ddot{m}_2}$ is positive because years when $\ddot{m}_2$ is high are years when matriculation probabilities are high in general. Using inter-year variation, this relationship results in a greater expected $\overline{m}_{2A}$ in those years. Then:

$$\begin{aligned} F_{\overline{m}_{2A}|\ddot{m}_2}(\overline{m}_{2A}|\ddot{m}_2^H;\beta_{\ddot{m}_2},\sigma^2_{\ddot{m}_2}) &= \Phi\left(\frac{\overline{m}_{2A} - \ddot{m}_2^H\beta_{\ddot{m}_2}}{\sigma_{\ddot{m}_2}}\right) \\ &> \Phi\left(\frac{\overline{m}_{2A} - \ddot{m}_2^L\beta_{\ddot{m}_2}}{\sigma_{\ddot{m}_2}}\right) = F_{\overline{m}_{2A}|\ddot{m}_2}(\overline{m}_{2A}|\ddot{m}_2^L;\beta_{\ddot{m}_2},\sigma^2_{\ddot{m}_2}). \end{aligned} \tag{11}$$

Therefore, $\overline{m}_{2A}|\ddot{m}_2^L$ first-order stochastically dominates $\overline{m}_{2A}|\ddot{m}_2^H$. To apply this stochastic dominance, I must show that $\sigma\log\left(e^{P(y=S|x,z,t,\bar{k})/\sigma} + e^{P(y_3^*=S|\overline{m}_{2A})/\sigma}\right)$ is strictly increasing in $\overline{m}_{2A}$. But the only place $\overline{m}_{2A}$ appears in this expression is in $P(y_3^* = S|\overline{m}_{2A};\theta_{S_3^*}) = \frac{\exp\left(f_{S_3^*}(\overline{m}_{2A})\theta_{S_3^*}\right)}{1+\exp\left(f_{S_3^*}(\overline{m}_{2A})\theta_{S_3^*}\right)}$, where by assumption $f_{S_3^*}(\cdot)$ is strictly increasing and $\theta_{S_3^*} > 0$. $\theta_{S_3^*}$ is positive because the expected Round 3 cutoff probability is higher when early acceptees are more likely on average to matriculate. As such, I have that $P(y_3^* = S|\overline{m}_{2A})$, and in turn $\sigma\log\left(e^{P(y=S|x,z,t,\bar{k})/\sigma} + e^{P(y_3^*=S|\overline{m}_{2A})/\sigma}\right)$, are strictly

increasing in $\overline{m}_{2A}$. Combining this result with $(\overline{m}_{2A}|\ddot{m}_2^L) \succ_{\text{FOSD}} (\overline{m}_{2A}|\ddot{m}_2^H)$, I obtain:

$$\begin{aligned}
&\mathbb{E}[v_3(x^L, z^L, t, \bar{k}, \overline{m}_{2A}, \varepsilon_3)|x^L, z^L, t, \bar{k}, \ddot{m}_2^L] \\
&\quad = \int_{\widetilde{\overline{m}_{2A}}} \sigma \log\left(e^{P(y=S|x^L,z^L,t,\bar{k})/\sigma} + e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right) dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2^L) \\
&\quad = \int_{\widetilde{\overline{m}_{2A}}} \sigma \log\left(e^{P(y=S|x^H,z^H,t,\bar{k})/\sigma} + e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right) dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2^L) \qquad (12) \\
&\quad > \int_{\widetilde{\overline{m}_{2A}}} \sigma \log\left(e^{P(y=S|x^H,z^H,t,\bar{k})/\sigma} + e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right) dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2^H) \\
&\quad = \mathbb{E}[v_3(x^H, z^H, t, \bar{k}, \overline{m}_{2A}, \varepsilon_3)|x^H, z^H, t, \bar{k}, \ddot{m}_2^H].
\end{aligned}$$

The second equality holds since $P(y = S|x^H, z^H, t, \bar{k}) = P(y = S|x^L, z^L, t, \bar{k})$. Thus, by Equation (10) and Inequality (12), I have $P(d_2 = A|x^H, z^H, t, \bar{k}, \ddot{m}_2^H) > P(d_2 = A|x^L, z^L, t, \bar{k}, \ddot{m}_2^L)$. □

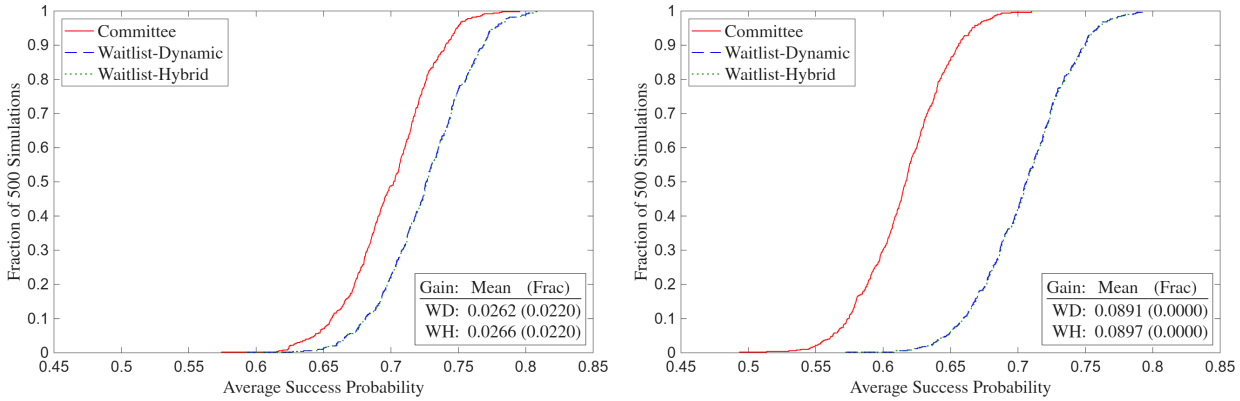
C.2 Asymmetric Waitlist Results

Unlike the two-round models, I do not provide estimation results for the waitlist models on actual data. The reason is that the relationship between $\overline{m_{2A}}$ and $\ddot{m}_2$ is only identified by variation across years, given that $\overline{m_{2A}}$ is constant within years. I have only five years of data, and the Covid shock, which may provide variation in $\overline{m_2}$ across years, is not part of the sample. As such, I have insufficient statistical power to estimate the F$\overline{m_{2A}}$ block of the likelihood, at least not until a potential future date when applicants from multiple application years after 2019 have completed their PhDs. Therefore, I stick to estimating the waitlist models on the same nearly optimal and suboptimal simulated datasets that I use to demonstrate that the deviation gains are considerably larger for the suboptimal than the optimal dataset. In the simulated data, there are only two years, but the data are generated such that the relationship between $\overline{m_{2A}}$ and $\ddot{m}_2$ is identified.[8] The purpose of this estimation on simulated data is to demonstrate that there exists a plausible data-generating process in which the waitlist models' parameters are identified.

The unrestricted waitlist-dynamic and hybrid estimates are in Tables 15 and 17. As expected, the $\ddot{m}_2$ parameter in the F$\overline{m_{2A}}$ block is positive and significant at the 1% level in both datasets. In addition, the $\ddot{m}_2$ parameter in the *Accept 2* block is negative and significant at the 1% level in both datasets, which is also as expected. The reason is that the committee is more likely to give an early offer to an applicant with a high m_2 and therefore a low $\ddot{m}_2$. Finally, the $\overline{m_{2A}}$ parameter in the *Accept 3* block is negative because a high $\overline{m_{2A}}$ means fewer spots will be open in Round 3 due to more Round 2 matriculants, and so it will be harder to get accepted off the waitlist.

Tables 16 and 18 contain restricted estimates for the waitlist-dynamic and hybrid models. In the waitlist-hybrid model, as with the two-round static model, the Round 1 cutoff probability is

[8] I set the weight that the committee puts on matriculation in Round 2 to be the same in both years, which keeps $\overline{m_{2A}}$ and $\ddot{m}_2$ strongly correlated. In the actual data, there is some variation in that weight across years, which along with only five years of data makes the correlation between $\overline{m_{2A}}$ and $\ddot{m}_2$ insignificantly different from zero.



(a) Optimal Committee

(b) Suboptimal Committee

Figure 6: Waitlist Deviation Gains (Matriculation Set)

higher relative to the Round 3 cutoff probability in the year that is not *Year Transition/Matriculate More*. In the suboptimal dataset, the marginal applicant rejected in Round 1's $P(y = S|x,t,\bar{k})$ is greater than the marginal applicant accepted into the program's $P(y = S|x,z,t,\bar{k})$, which may be undesirable for the committee. Also, in *Year Transition/Matriculate More*, the Round 3 cutoff probability is higher than the Round 1 cutoff probability not only because there are more Round 1 survivors, but also because those applicants have higher matriculation probabilities on average.

Finally, I plot the bootstrap-corrected deviation gains for the waitlist models. For the waitlist-dynamic model, I rank the applicants in each year by $\mathbb{E}[v_2(x,z,t,\bar{k},\ddot{m}_2,\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}} \sigma \log\left(e^{P(y=S|x,\tilde{z},t,\bar{k})/\sigma} + e^{\beta\mathbb{E}[v_3(x,\tilde{z},t,\bar{k},\widetilde{m_{2A}},\tilde{\varepsilon}_3)|x,\tilde{z},t,\bar{k},\ddot{m}_2(x,\tilde{z},t)]/\sigma}\right) dF_{z|x,t}(\tilde{z}|x,t)$ and transition the number of transitioned applicants in the data. Next, I rank the surviving applicants by the payoff difference $P(y = S|x,z,t,\bar{k}) - \beta\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m_{2A}},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2] = P(y = S|x,z,t,\bar{k}) - \beta\int_{\widetilde{m_{2A}}} \sigma \log\left(e^{P(y=S|x,z,t,\bar{k})/\sigma} + e^{P(y_3^*=S|\widetilde{m_{2A}})/\sigma}\right) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{m_{2A}}|\ddot{m}_2)$, accept the number of early acceptances in the data, and waitlist the rest. Each early acceptance matriculates with early matriculation probability $P(m_2 = M|x,z,t)$. After that, I rank the waitlisted applicants by $P(y = S|x,z,t,\bar{k})$ and accept them in descending order, each matriculating with late matriculation probability $P(m_3 = M|x,z,t)$, until the number of early plus late matriculants equals that in the data. Lastly, I compute the corrected average success probability of the matriculation set and compare it to the committee's set. For the waitlist-hybrid model, I instead rank the applicants in Round 1 by $P(y = S|x,t,\bar{k})$. Per Figure 6, the deviation gains across waitlist models are over three times as large in the suboptimal dataset as in the optimal dataset.

C.3 Symmetric Waitlist Models

In Appendix C.1, I present waitlist models in which success probabilities equal, the committee is more likely to give Round 2 acceptances to likely matriculants. Intuitively, this primary channel holds because likely matriculants decrease the expected number of spots to fill in Round 3, letting

Table 15: Simulated Unrestricted Waitlist-Dynamic Estimates

(a) Optimal Committee

	Advanced Math		Committee Score	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577**	-0.4303**	0.3695	-0.0077	-12.0690***	-2.2483***	-10.3023***	-1.5622***	-5.8123***	-0.7870***	-7.3172***	-1.0451***	9.2144	0.9184
	(0.7805)	(0.1891)	(1.4912)	(0.0182)	(0.9758)	(0.1337)	(2.0855)	(0.2950)	(2.1668)	(0.2925)	(1.2885)	(0.1661)	(7.0351)	(0.6914)
GRE Quantative Percentile	0.0251***	0.0061***	0.0510***		0.1260***	0.0235***	0.1065***	0.0162***	0.0513**	0.0069**	0.1028***	0.0147***	-0.0982	-0.0098
	(0.0094)	(0.0023)	(0.0170)		(0.0114)	(0.0017)	(0.0235)	(0.0034)	(0.0244)	(0.0033)	(0.0142)	(0.0017)	(0.0729)	(0.0072)
Gender	-0.2435*	-0.0596*	0.1807		0.0400	0.0074	0.1268	0.0192	-0.0757	-0.0102	0.2259	0.0323	-0.5262	-0.0524
	(0.1312)	(0.0319)	(0.2034)		(0.1525)	(0.0284)	(0.2763)	(0.0417)	(0.3354)	(0.0455)	(0.1762)	(0.0251)	(0.7680)	(0.0749)
Demographic 1	-0.0504	-0.0123	-0.6989**		0.0163	0.0030	-0.2346	-0.0356	0.1428	0.0193	-0.4384*	-0.0626*	1.8221*	0.1816*
	(0.1730)	(0.0424)	(0.2889)		(0.2031)	(0.0378)	(0.4296)	(0.0649)	(0.5011)	(0.0677)	(0.2358)	(0.0334)	(1.0613)	(0.1030)
Demographic 2	-0.3121	-0.0764	-0.4226		0.1678	0.0313	0.0460	0.0070	-0.6064	-0.0821	-0.4682*	-0.0669*	0.3786	0.0377
	(0.1930)	(0.0470)	(0.3254)		(0.2315)	(0.0431)	(0.4601)	(0.0698)	(0.6255)	(0.0845)	(0.2734)	(0.0387)	(1.4349)	(0.1440)
Advanced Math			1.1347***				0.1026	0.0156	0.2643	0.0358	0.7127***	0.1018***	1.3289*	0.1324*
			(0.1992)				(0.2866)	(0.0434)	(0.3393)	(0.0457)	(0.1791)	(0.0247)	(0.7058)	(0.0702)
Committee Score							0.0267	0.0041	0.0955	0.0129	-0.0790	-0.0113	-1.0874***	-0.1084***
							(0.0752)	(0.0114)	(0.0902)	(0.0122)	(0.0688)	(0.0098)	(0.3076)	(0.0197)
Missing Committee Score											0.0486	0.0069		
											(0.1969)	(0.0281)		
Year Transition/Matriculate More	-0.2523**	-0.0618**	0.1483		1.1441***	0.2131***					-0.3543**	-0.0506**	5.0805***	0.5064***
	(0.1282)	(0.0311)	(0.2025)		(0.1501)	(0.0250)					(0.1794)	(0.0254)	(0.1012)	(0.0379)
Program Rank											-0.0302***	-0.0043***		
											(0.0018)	(0.0002)		
$\ddot{m}_2$				0.9426***			-1.4274***	-0.2164***						
				(0.0721)			(0.4888)	(0.0725)						
$\overline{m_{2A}}$									-1.9017***	-0.2575***				
									(0.6371)	(0.0837)				
$\sqrt{\text{MSE}}$			1.8393***	0.0013***										
			(0.0614)	(0.0002)										
Obs | Param | LL | AIC	1000	49	-800.8	1699.7										

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Advanced Math		Committee Score	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.7577**	-0.4303**	1.2187	-0.0002	-10.3454***	-2.0151***	-7.3844***	-1.1920***	-2.7175	-0.4361	-7.3017***	-1.0448***	2.5184	0.2378
	(0.7805)	(0.1891)	(1.4016)	(0.0096)	(0.9474)	(0.1440)	(1.8422)	(0.2834)	(1.9623)	(0.3142)	(1.2721)	(0.1635)	(5.5971)	(0.5349)
GRE Quantative Percentile	0.0251***	0.0061***	0.0431***		0.1103***	0.0215***	0.0749***	0.0121***	0.0083	0.0013	0.1021***	0.0146***	-0.0190	-0.0018
	(0.0094)	(0.0023)	(0.0164)		(0.0111)	(0.0018)	(0.0214)	(0.0033)	(0.0236)	(0.0038)	(0.0140)	(0.0017)	(0.0620)	(0.0059)
Gender	-0.2435*	-0.0596*	0.1707		-0.0472	-0.0092	0.3247	0.0524	-0.0166	-0.0027	0.2087	0.0299	-0.4992	-0.0471
	(0.1312)	(0.0319)	(0.2037)		(0.1485)	(0.0289)	(0.2679)	(0.0428)	(0.3210)	(0.0515)	(0.1755)	(0.0251)	(0.7149)	(0.0664)
Demographic 1	-0.0504	-0.0123	-0.7422***		-0.4655**	-0.0907**	-0.9396**	-0.1517**	0.0039	0.0006	-0.4120*	-0.0590*	1.2276	0.1159
	(0.1730)	(0.0424)	(0.2732)		(0.1978)	(0.0382)	(0.3764)	(0.0592)	(0.4412)	(0.0708)	(0.2351)	(0.0334)	(0.8815)	(0.0867)
Demographic 2	-0.3121	-0.0764	-0.4868*		0.1277	0.0249	-0.2162	-0.0349	-0.0882	-0.0142	-0.3798	-0.0543	1.3234	0.1249
	(0.1930)	(0.0470)	(0.2925)		(0.2186)	(0.0426)	(0.3862)	(0.0623)	(0.4770)	(0.0765)	(0.2696)	(0.0384)	(0.9828)	(0.0937)
Advanced Math			1.1482***				-0.0885	-0.0143	0.0454	0.0073	0.7178***	0.1027***	-0.2675	-0.0253
			(0.1999)				(0.2806)	(0.0453)	(0.3151)	(0.0505)	(0.1795)	(0.0248)	(0.6906)	(0.0638)
Committee Score							0.1489**	0.0240**	0.1649**	0.0265**	-0.0769	-0.0110	-0.9443***	-0.0892***
							(0.0755)	(0.0120)	(0.0836)	(0.0133)	(0.0684)	(0.0098)	(0.2437)	(0.0194)
Missing Committee Score											0.1129	0.0162		
											(0.1909)	(0.0273)		
Year Transition/Matriculate More	-0.2523**	-0.0618**	0.0871		1.0879***	0.2119***					-0.3741**	-0.0535**	5.6972***	0.5379***
	(0.1282)	(0.0311)	(0.2014)		(0.1466)	(0.0255)					(0.1781)	(0.0253)	(0.9551)	(0.0235)
Program Rank											-0.0309***	-0.0044***		
											(0.0019)	(0.0002)		
$\ddot{m}_2$				0.9096***			-1.4175***	-0.2288***						
				(0.0363)			(0.3868)	(0.0591)						
$\overline{m_{2A}}$									-0.3921	-0.0629				
									(0.5053)	(0.0811)				
$\sqrt{\text{MSE}}$			1.8362***	0.0010***										
			(0.0603)	(0.0002)										
Obs | Param | LL | AIC	1000	49	-735.6	1569.3										

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 16: Simulated Restricted Waitlist-Dynamic Estimates

(a) Optimal Committee

	Advanced Math	Committee Score	$\overline{m_{2A}}$	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success	Matriculate	Beta	Sigma
Intercept	-1.9385**	0.3755	-0.0077	-0.0462	0.9548	0.7613**	70.31%	-7.7403***	9.2144		
	(0.7791)	(1.4941)	(0.0182)	(0.0706)	(0.8314, 1.0965)	(0.3224)	(57.57%, 80.51%)	(1.1291)	(7.0351)		
GRE Quantative Percentile	0.0272***	0.0509***						0.1037***	-0.0982		
	(0.0094)	(0.0171)						(0.0129)	(0.0729)		
Gender	-0.2476*	0.1808						0.0942	-0.5263		
	(0.1310)	(0.2034)						(0.1161)	(0.7680)		
Demographic 1	-0.0434	-0.6992**						-0.2037	1.8221*		
	(0.1727)	(0.2903)						(0.1559)	(1.0613)		
Demographic 2	-0.3031	-0.4230						-0.2725	0.3786		
	(0.1930)	(0.3260)						(0.1818)	(1.4349)		
Advanced Math		1.1347***						0.4532***	1.3289*		
		(0.1992)						(0.1448)	(0.7058)		
Committee Score								-0.0017	-1.0874***		
								(0.0431)	(0.3076)		
Missing Committee Score								0.0156			
								(0.1893)			
Year Transition/Matriculate More	-0.2451*	0.1482		-0.1522***	0.8200		80.07%	-0.3714**	5.0805***		
	(0.1281)	(0.2023)		(0.0558)	(0.7168, 0.9381)		(66.04%, 89.24%)	(0.1763)	(1.0112)		
Program Rank								-0.0300***			
								(0.0018)			
$\dot{m}_2$			0.9426***								
			(0.0721)								
$\overline{m_{2A}}$						0.9566					
						(0.6998)					
$\sqrt{\text{MSE}}$		1.8393***	0.0013***								
		(0.0614)	(0.0002)								
Beta										0.9372***	
										(0.0376)	
Sigma											0.1631***
											(0.0243)
Obs | Param | LL | AIC	1000	33	-844.5	1755.0							

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Advanced Math	Committee Score	$\overline{m_{2A}}$	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success	Matriculate	Beta	Sigma
Intercept	-1.9232**	1.0234	-0.0002	0.0583	1.0601	0.9615**	73.34%	-7.5884***	2.5184		
	(0.7773)	(1.4051)	(0.0096)	(0.1006)	(0.8704, 1.2911)	(0.3901)	(57.37%, 84.90%)	(1.1982)	(5.5971)		
GRE Quantative Percentile	0.0271***	0.0454***						0.0994***	-0.0190		
	(0.0093)	(0.0165)						(0.0135)	(0.0620)		
Gender	-0.2510*	0.1637						0.1395	-0.4992		
	(0.1310)	(0.2039)						(0.1293)	(0.7149)		
Demographic 1	-0.0530	-0.7463***						-0.5213***	1.2276		
	(0.1728)	(0.2729)						(0.1677)	(0.8815)		
Demographic 2	-0.3009	-0.4761						-0.2142	1.3234		
	(0.1928)	(0.2928)						(0.1987)	(0.9828)		
Advanced Math		1.1476***						0.4353***	-0.2675		
		(0.1999)						(0.1595)	(0.6906)		
Committee Score								0.0640	-0.9443***		
								(0.0471)	(0.2437)		
Missing Committee Score								0.0867			
								(0.1864)			
Year Transition/Matriculate More	-0.2468*	0.0909		-0.2298***	0.8424		79.86%	-0.3802**	5.6972***		
	(0.1281)	(0.2010)		(0.0592)	(0.6960, 1.0197)		(60.97%, 90.96%)	(0.1755)	(0.9551)		
Program Rank								-0.0303***			
								(0.0018)			
$\dot{m}_2$			0.9096***								
			(0.0363)								
$\overline{m_{2A}}$						0.5686					
						(0.7016)					
$\sqrt{\text{MSE}}$		1.8362***	0.0010***								
		(0.0603)	(0.0002)								
Beta										0.9121***	
										(0.0444)	
Sigma											0.2220***
											(0.0415)
Obs | Param | LL | AIC	1000	33	-781.3	1628.6							

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 17: Simulated Unrestricted Waitlist-Hybrid Estimates

(a) Optimal Committee

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0077	-12.0690***	-2.2483***	-10.3023***	-1.5622***	-5.8123***	-0.7870***	-7.3875***	-1.0821***	-7.3172***	-1.0451***	9.2144	0.9184
	(0.0182)	(0.9758)	(0.1337)	(2.0855)	(0.2950)	(2.1668)	(0.2925)	(1.1197)	(0.1406)	(1.2885)	(0.1661)	(7.0351)	(0.6914)
GRE Quantitative Percentile		0.1260***	0.0235***	0.1065***	0.0162***	0.0513**	0.0069**	0.1043***	0.0153***	0.1028***	0.0147***	-0.0982	-0.0098
		(0.0114)	(0.0017)	(0.0235)	(0.0034)	(0.0244)	(0.0033)	(0.0134)	(0.0016)	(0.0142)	(0.0017)	(0.0729)	(0.0072)
Gender		0.0400	0.0074	0.1268	0.0192	-0.0757	-0.0102	0.1580	0.0231	0.2259	0.0323	-0.5262	-0.0524
		(0.1525)	(0.0284)	(0.2763)	(0.0417)	(0.3354)	(0.0455)	(0.1732)	(0.0254)	(0.1762)	(0.0251)	(0.7680)	(0.0749)
Demographic 1		0.0163	0.0030	-0.2346	-0.0356	0.1428	0.0193	-0.4556**	-0.0667**	-0.4384*	-0.0626*	1.8221*	0.1816*
		(0.2031)	(0.0378)	(0.4296)	(0.0649)	(0.5011)	(0.0677)	(0.2269)	(0.0328)	(0.2358)	(0.0334)	(1.0613)	(0.1030)
Demographic 2		0.1678	0.0313	0.0460	0.0070	-0.6064	-0.0821	-0.5153*	-0.0755**	-0.4682*	-0.0669*	0.3786	0.0377
		(0.2315)	(0.0431)	(0.4601)	(0.0698)	(0.6255)	(0.0845)	(0.2641)	(0.0383)	(0.2734)	(0.0387)	(1.4349)	(0.1440)
Advanced Math				0.1026	0.0156	0.2643	0.0358			0.7127***	0.1018***	1.3289*	0.1324*
				(0.2866)	(0.0434)	(0.3393)	(0.0457)			(0.1791)	(0.0247)	(0.7058)	(0.0702)
Committee Score				0.0267	0.0041	0.0955	0.0129			-0.0790	-0.0113	-1.0874***	-0.1084***
				(0.0752)	(0.0114)	(0.0902)	(0.0122)			(0.0688)	(0.0098)	(0.3076)	(0.0197)
Missing Committee Score										0.0486	0.0069		
										(0.1969)	(0.0281)		
Year Transition/Matriculate More		1.1441***	0.2131***					-0.3942**	-0.0577**	-0.3543**	-0.0506**	5.0805***	0.5064***
		(0.1501)	(0.0250)					(0.1689)	(0.0244)	(0.1794)	(0.0254)	(1.0112)	(0.0379)
Program Rank								-0.0299***	-0.0044***	-0.0302***	-0.0043***		
								(0.0018)	(0.0002)	(0.0018)	(0.0002)		
$\dot{m}_2$	0.9426***			-1.4274***	-0.2164***								
	(0.0721)			(0.4888)	(0.0725)								
$\overline{m_{2A}}$						-1.9017***	-0.2575***						
						(0.6371)	(0.0837)						
$\sqrt{\text{MSE}}$	0.0013***												
	(0.0002)												
Obs | Param | LL | AIC	1000	42	169.4	-254.7									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0002	-10.3454***	-2.0151***	-7.3844***	-1.1920***	-2.7175	-0.4361	-7.2362***	-1.0632***	-7.3017***	-1.0448***	2.5184	0.2378
	(0.0096)	(0.9474)	(0.1440)	(1.8422)	(0.2834)	(1.9623)	(0.3142)	(1.1209)	(0.1418)	(1.2721)	(0.1635)	(5.5971)	(0.5349)
GRE Quantitative Percentile		0.1103***	0.0215***	0.0749***	0.0121***	0.0083	0.0013	0.1026***	0.0151***	0.1021***	0.0146***	-0.0190	-0.0018
		(0.0111)	(0.0018)	(0.0214)	(0.0033)	(0.0236)	(0.0038)	(0.0134)	(0.0016)	(0.0140)	(0.0017)	(0.0620)	(0.0059)
Gender		-0.0472	-0.0092	0.3247	0.0524	-0.0166	-0.0027	0.1408	0.0207	0.2087	0.0299	-0.4992	-0.0471
		(0.1485)	(0.0289)	(0.2679)	(0.0428)	(0.3210)	(0.0515)	(0.1727)	(0.0254)	(0.1755)	(0.0251)	(0.7149)	(0.0664)
Demographic 1		-0.4655**	-0.0907**	-0.9396**	-0.1517**	0.0039	0.0006	-0.4199*	-0.0617*	-0.4120*	-0.0590*	1.2276	0.1159
		(0.1978)	(0.0382)	(0.3764)	(0.0592)	(0.4412)	(0.0708)	(0.2285)	(0.0332)	(0.2351)	(0.0334)	(0.8815)	(0.0867)
Demographic 2		0.1277	0.0249	-0.2162	-0.0349	-0.0882	-0.0142	-0.4343*	-0.0638*	-0.3798	-0.0543	1.3234	0.1249
		(0.2186)	(0.0426)	(0.3862)	(0.0623)	(0.4770)	(0.0765)	(0.2605)	(0.0380)	(0.2696)	(0.0384)	(0.9828)	(0.0937)
Advanced Math				-0.0885	-0.0143	0.0454	0.0073			0.7178***	0.1027***	-0.2675	-0.0253
				(0.2806)	(0.0453)	(0.3151)	(0.0505)			(0.1795)	(0.0248)	(0.6906)	(0.0638)
Committee Score				0.1489**	0.0240**	0.1649**	0.0265**			-0.0769	-0.0110	-0.9443***	-0.0892***
				(0.0755)	(0.0120)	(0.0836)	(0.0133)			(0.0684)	(0.0098)	(0.2437)	(0.0194)
Missing Committee Score										0.1129	0.0162		
										(0.1909)	(0.0273)		
Year Transition/Matriculate More		1.0879***	0.2119***					-0.4350**	-0.0639***	-0.3741**	-0.0535**	5.6972***	0.5379***
		(0.1466)	(0.0255)					(0.1689)	(0.0244)	(0.1781)	(0.0253)	(0.9551)	(0.0235)
Program Rank								-0.0305***	-0.0045***	-0.0309***	-0.0044***		
								(0.0019)	(0.0002)	(0.0019)	(0.0002)		
$\dot{m}_2$	0.9096***			-1.4175***	-0.2288***								
	(0.0363)			(0.3868)	(0.0591)								
$\overline{m_{2A}}$						-0.3921	-0.0629						
						(0.5053)	(0.0811)						
$\sqrt{\text{MSE}}$	0.0010***												
	(0.0002)												
Obs | Param | LL | AIC	1000	42	239.8	-395.5									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 18: Simulated Restricted Waitlist-Hybrid Estimates

(a) Optimal Committee

	$\overline{m_{2A}}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	-0.0077	1.2144***	77.11%	1.0190**	77.53%	-7.8124***	-6.9318***	9.2144		
	(0.0182)	(0.2511)	(67.31%, 84.64%)	(0.4475)	(60.50%, 88.59%)	(1.0695)	(1.2242)	(7.0351)		
GRE Quantitative Percentile						0.1066***	0.0956***	-0.0982		
						(0.0128)	(0.0139)	(0.0729)		
Gender						0.0695	0.1380	-0.5263		
						(0.1079)	(0.1368)	(0.7680)		
Demographic 1						-0.1643	-0.3059	1.8221*		
						(0.1412)	(0.1938)	(1.0613)		
Demographic 2						-0.1000	-0.3932*	0.3786		
						(0.1626)	(0.2184)	(1.4349)		
Advanced Math							0.5098***	1.3289*		
							(0.1453)	(0.7058)		
Committee Score							-0.0100	-1.0874***		
							(0.0455)	(0.3076)		
Missing Committee Score							-0.0160			
							(0.1924)			
Year Transition/Matriculate More		-1.3719***	46.07%		91.60%	-0.3711**	-0.3826**	5.0805***		
		(0.2745)	(40.01%, 52.25%)		(64.02%, 98.53%)	(0.1649)	(0.1814)	(1.0112)		
Program Rank						-0.0298***	-0.0300***			
						(0.0018)	(0.0018)			
$\dot{m}_2$	0.9426***									
	(0.0721)									
$\overline{m_{2A}}$				2.0829						
				(1.5159)						
$\sqrt{\text{MSE}}$	0.0013***									
	(0.0002)									
Beta									0.9120***	
									(0.0387)	
Sigma										0.2026***
										(0.0253)
Obs | Param | LL | AIC	1000	26	159.6	-267.3						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	$\overline{m_{2A}}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	-0.0002	1.5000***	81.76%	1.3822**	81.24%	-7.9054***	-6.3534***	2.5184		
	(0.0096)	(0.3795)	(68.05%, 90.41%)	(0.5676)	(60.16%, 92.54%)	(1.1157)	(1.2856)	(5.5971)		
GRE Quantitative Percentile						0.1100***	0.0859***	-0.0190		
						(0.0133)	(0.0144)	(0.0620)		
Gender						0.0337	0.2012	-0.4992		
						(0.1205)	(0.1425)	(0.7149)		
Demographic 1						-0.4489***	-0.5069***	1.2276		
						(0.1584)	(0.1908)	(0.8815)		
Demographic 2						-0.1162	-0.3213	1.3234		
						(0.1792)	(0.2184)	(0.9828)		
Advanced Math							0.4795***	-0.2675		
							(0.1542)	(0.6906)		
Committee Score							0.0529	-0.9443***		
							(0.0486)	(0.2437)		
Missing Committee Score							0.0335			
							(0.1912)			
Year Transition/Matriculate More		-1.6732***	45.68%		88.77%	-0.4035**	-0.4145**	5.6972***		
		(0.3924)	(39.02%, 52.50%)		(60.45%, 97.61%)	(0.1655)	(0.1788)	(0.9551)		
Program Rank						-0.0303***	-0.0305***			
						(0.0019)	(0.0018)			
$\dot{m}_2$	0.9096***									
	(0.0363)									
$\overline{m_{2A}}$				0.9360						
				(1.1799)						
$\sqrt{\text{MSE}}$	0.0010***									
	(0.0002)									
Beta									0.8893***	
									(0.0434)	
Sigma										0.2563***
										(0.0392)
Obs | Param | LL | AIC	1000	26	218.0	-384.0						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

the committee be more selective. But there is also a secondary channel: if the committee accepts likely matriculants in Round 2, then the unlikely matriculants it waitlists end up in the Round 3 pool. As such, while there are fewer expected spots to fill, it is easier to fill those spots. For tractability, I do not estimate this secondary channel in the main specification. But tractability aside, there are two economic reasons why the primary channel is dominant. First, matriculation probabilities decline between Rounds 2 and 3, and for likely matriculants, they have more room to fall. Second, matriculation probabilities can change unpredictably between those rounds.

Still, I present symmetric waitlist models that account for both the primary and secondary channels. There are two additional states: $\overline{m_3}$, which is a given year's average waitlisted applicant matriculation probability, and m_3, which is the *ex-ante* probability that an applicant will matriculate if waitlisted. By *ex-ante*, I mean that m_3 depends only on information in the application files and not events that happen between Rounds 2 and 3. Therefore, the committee knows m_3 in Round 2. As with $\overline{m_{2A}}$, the Round 3 cutoff probability is increasing in $\overline{m_3}$. In addition, $\overline{m_3}$ is increasing in m_3, analogously to how $\overline{m_{2A}}$ is increasing in $\ddot{m}_2$. Whether the committee will be more or less likely to accept likely matriculants in Round 2 is theoretically ambiguous and depends on the parameter values. The equations are as follows, starting with Round 3:

$$v_3(x,z,t,\bar{k},\overline{m_{2A}},\overline{m_3},\varepsilon_3) = \max_{d_3\in\{A,R\}} u_3(x,z,t,\bar{k},\overline{m_{2A}},\overline{m_3},d_3)+\varepsilon_3(d_3), \text{ where:} \tag{13}$$

$$u_3(x,z,t,\bar{k},\overline{m_{2A}},\overline{m_3},d_3) = P(y=S|x,z,t,\bar{k})\mathbb{1}(d_3=A)+P(y_3^*=S|\overline{m_{2A}},\overline{m_3})\mathbb{1}(d_3=R).$$

The Round 2 value function is:

$$v_2(x,z,t,\bar{k},\ddot{m}_2,m_3,\varepsilon_2) = \max_{d_2\in\{A,W\}} u_2(x,z,t,\bar{k},\ddot{m}_2,m_3,d_2)+\varepsilon_2(d_2), \text{ where:} \tag{14}$$

$$\begin{aligned} u_2(x,z,t,\bar{k},\ddot{m}_2,m_3,d_2) &= P(y=S|x,z,t,\bar{k})\mathbb{1}(d_2=A) \\ &\quad +\beta\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m_{2A}},\overline{m_3},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2,m_3]\mathbb{1}(d_2=W), \text{ where:} \end{aligned}$$

$$\begin{aligned} &\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m_{2A}},\overline{m_3},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2,m_3] \\ &= \int_{\widetilde{\overline{m_{2A}}}}\int_{\widetilde{\overline{m_3}}} \sigma\log\left(\sum_{\widetilde{d_3}\in\{A,R\}} e^{u_3(x,z,t,\bar{k},\widetilde{\overline{m_{2A}}},\widetilde{\overline{m_3}},\widetilde{d_3})/\sigma}\right) dF_{\overline{m_3}|m_3}(\widetilde{\overline{m_3}}|m_3)\,dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2). \end{aligned}$$

Finally, the Round 1 value function is:

$$v_1(x,t,\bar{k},\varepsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,\bar{k},d_1)+\varepsilon_1(d_1), \text{ where:} \tag{15}$$

$$u_1(x,t,\bar{k},d_1) = \mathbb{E}[v_2(x,z,t,\bar{k},\ddot{m}_2,m_3,\varepsilon_2)|x,t,\bar{k}]\mathbb{1}(d_1=T)+L(y_1^*=S|t)\mathbb{1}(d_1=R), \text{ where:}$$

$$\mathbb{E}[v_2(x,z,t,\bar{k},\ddot{m}_2,m_3,\varepsilon_2)|x,t,\bar{k}] = \int_{\widetilde{z}}\sigma\log\left(\sum_{\widetilde{d_2}\in\{A,W\}} e^{u_2(x,\widetilde{z},t,\bar{k},\ddot{m}_2(x,\widetilde{z},t),m_3(x,\widetilde{z},t),\widetilde{d_2})/\sigma}\right) dF_{z|x,t}(\widetilde{z}|x,t).$$

In the symmetric waitlist-hybrid model, the Round 1 utility function is instead:

$$u_1(x,t,\bar{k},d_1) = P(y=S|x,t,\bar{k})\mathbb{1}(d_1=T) + P(y_1^*=S|t)\mathbb{1}(d_1=R). \tag{16}$$

Expressed as simply as possible, the restricted CCPs are:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states},d_r)/\sigma)}{\sum_{\widetilde{d}_r\in D_r}\exp(u_r(\text{states},\widetilde{d}_r)/\sigma)}. \tag{17}$$

The unrestricted CCPs, success probabilities, and cutoff level/probability are $P_{d_1} = P\big(d_1 = T\big|x,t;\theta_T\big)$, $P_{d_2} = P\big(d_2 = A\big|x,z,\ddot{m_2},m_3;\theta_{A_2}\big)$, $P_{d_3} = P\big(d_3 = A\big|x,z,\overline{m_{2A}},\overline{m_3};\theta_{A_3}\big)$, $P_y = P\big(y = S\big|x,z,t,\bar{k};\theta_S\big)$, $L_{y_1^*} = L\big(y_1^* = S\big|t;\theta_{S_1^*}\big)$, and $P_{y_3^*} = P\big(y_3^* = S\big|\overline{m_{2A}},\overline{m_3};\theta_{S_3^*}\big)$. They have the same specifications as the probabilities and levels earlier in the paper; see Equations (11)–(13) in Section 4.1.1. Let applicants 1 to N_W be waitlisted, N_W+1 to N_T be accepted in Round 2, and N_T+1 to N be rejected in Round 1. Then the unrestricted symmetric waitlist-dynamic likelihood is:

$$\begin{aligned}\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} f\big(z_n^c|z_n^{\neg c},x_n,t_n;\theta_{F_z}\big) f\big(\overline{m_{2A,n}}|\ddot{m_{2,n}};\theta_{F_{m_2}}\big) f\big(\overline{m_{3,n}}|m_{3,n};\theta_{F_{m_3}}\big) \\ P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_{A_2})^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_{A_2}))^{\mathbb{1}(d_{2,n}=W)} \prod_{n=1}^{N_W} P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}(1-P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)} \\ \prod_{n=N_T+1}^{N}\big(1-P_{d_{1,n}}(\theta_T)\big)\prod_{n=1}^{N} f\big(z_n^{\neg c}|x_n,t_n;\theta_{F_z}\big) P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}.\end{aligned} \tag{18}$$

In addition, m_2 and m_3 are parameterized as follows: $m_2 = P_{m_2} = P\big(m_2 = M\big|x,z,t;\theta_{M_2}\big)$ and $m_3 = P_{m_3} = P\big(m_3 = M\big|x,z,t;\theta_{M_3}\big)$. To estimate θ_{M_2} and θ_{M_3}, I would use the first-stage likelihoods: $\mathcal{L}_{N_T-N_W}^{M_2}(\theta_{M_2}) = \prod_{n=N_W+1}^{N_T} P_{m_{2,n}}(\theta_{M_2})^{\mathbb{1}(m_{2,n}=M)}(1-P_{m_{2,n}}(\theta_{M_2}))^{\mathbb{1}(m_{2,n}=D)}$ and $\mathcal{L}_{N_W}^{M_3}(\theta_{M_3}) = \prod_{n=1}^{N_W} P_{m_{3,n}}(\theta_{M_3})^{\mathbb{1}(d_{3,n}=A,m_{3,n}=M)}(1-P_{m_{3,n}}(\theta_{M_3}))^{\mathbb{1}(d_{3,n}=A,m_{3,n}=D)}$. The samples for estimating m_2 and m_3 are applicants accepted in Rounds 2 and 3 respectively. Since m_2 and m_3 are generated regressors, the standard errors are those from Murphy and Topel (1985) and Hardin (2002).

Finally, in Appendix D, I present two different infinite-horizon multiyear models. A limitation of the waitlist models is that while they give early acceptances to applicants with higher matriculation probabilities all else equal, matriculation probabilities do not affect the probability of a late acceptance. For this reason, the first multiyear model is constructed so that an applicant's overall acceptance probability is increasing in their matriculation probability, the logic being that more matriculants this year will give the committee the option to be more selective next year. Meanwhile, the second multiyear model rewards more diverse applicants in a scenario where diversity increases an applicant's success probability. For example, by accepting more theory applicants this year, the committee may have to worry less next year about having too many applied students relative to applied faculty advisors. But this is just one example of such a scenario.

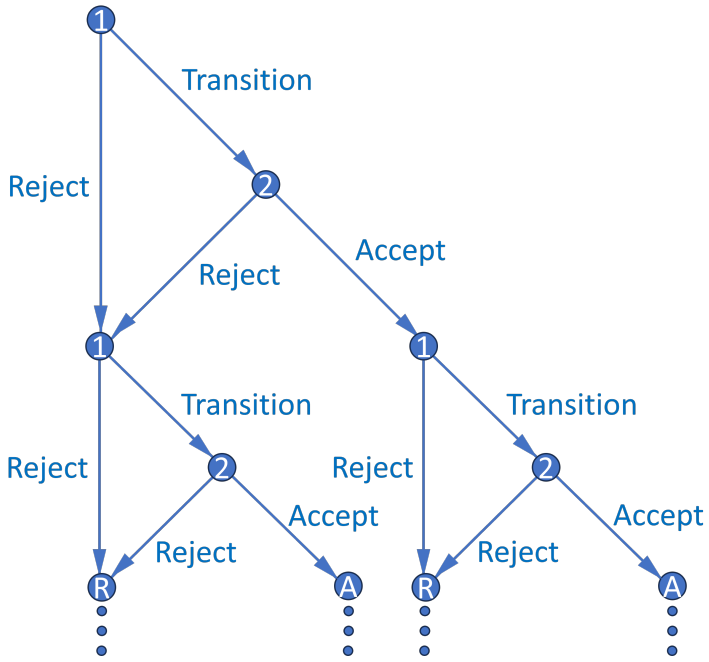


Figure 7: Multiyear Decision Graph

D Multiyear Models

There are two different multiyear models. In the first model, the memory variable is the size of the previous year's cohort. The committee accepts applicants more likely to matriculate because it wants to ensure that next year's memory variable, which is this year's cohort size, is sufficiently large that it can avoid accepting too many relatively weak applicants next year. In the second model, the memory is a diversity variable that increases an applicant's success probability; for example, the previous year's cohort's main area of interest. Applicants with a diverse area of interest may be more likely to succeed because their advisor, with fewer other students to advise, can give them more individualized attention. The committee will therefore accept applicants with a less common area of interest to help balance out the program for next year's class.

D.1 Matriculation Model

In the waitlist models in Appendix C, the committee benefits from giving early acceptances to applicants who are more likely to matriculate. Doing so helps prevent the program from losing out on good applicants who would likely have matriculated if given an early offer but matriculate elsewhere due to events that occurred while on the waitlist. However, the waitlist models do not imply that the committee should also give more late acceptances to likely matriculants. To model that phenomenon, a multiyear model is necessary, as there must be a period after the final round in any given year for dynamics to matter in that round. Figure 7 is a decision graph of the simplest possible multiyear model, in which each year has only two rounds, as in the dynamic model.

This model is infinite horizon, and future years are discounted with factor $\beta \in (0,1)$. To avoid the considerations of two-stage estimation, I suppose for simplicity that matriculation probabilities m come from the committee's assessment of each applicant in Round 2. I also suppose that success probabilities depend only on applicant characteristics and not program rank.

In each year's Round 2 value function, I specify the cutoff probability as $P(y_2^* = S|c_{-1})$ instead of $P(y_2^* = S|t)$, where c_{-1} is the previous year's cohort size. The cutoff probability increases in c_{-1} because the committee must keep the total number of students in the program across years relatively constant. So if c_{-1} is large, the committee must now accept fewer students.

Given that, what information available at the start of Round 2 determines c? If and only if the applicant is accepted, their own matriculation probability m does. Now suppose the committee maximizes discounted "lifetime" cohort success by solving an infinite-horizon dynamic programming problem across years. As in the waitlist models, the optimal dynamic decision may not match the optimal myopic one. Instead, the committee may accept an applicant with a high m even if their expected success probability is less than what the Round 2 cutoff level (which need not be between 0 and 1) would otherwise be. The reason is that accepting a high m applicant increases c's distribution, which increases next year's Round 2 cutoff and value.

Intuitively, even if the committee can accept more applicants next year to make up for a smaller cohort this year, they will do so out of necessity, for example, to avoid future TA shortages. This situation will be undesirable provided next year's marginal applicant is of lower quality than this year's above-marginal applicant. Also, as in the waitlist models, even if the data show that the committee does not think dynamically, if c_{-1} affects the Round 2 cutoff, perhaps it should. Finally, estimating this model is more computationally challenging since it does not have a closed-form solution. Nested fixed point algorithms suffer from the curse of dimensionality, and the number of state variables in application files, in particular those from textual documents, is large. But principal component analysis can reduce the size of the state space.

I now present the value functions, starting with each year's Round 2 value:

$$v_2(x,z,t,m,c_{-1},\varepsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x,z,t,m,c_{-1},d_2) + \varepsilon_2(d_2), \text{ where:} \tag{19}$$

$$\begin{aligned} u_2(x,z,t,m,c_{-1},d_2) = &\{P(y=S|x,z,t) + \beta \mathbb{E}[v_1(x',t',c,\varepsilon_1{}')|m]\}\mathbb{1}(d_2 = A) \\ &+ \{L(y_2^* = S|c_{-1}) + \beta \mathbb{E}[v_1(x',t',c,\varepsilon_1{}')]\}\mathbb{1}(d_2 = R), \text{ where:} \end{aligned}$$

$$\begin{aligned} \mathbb{E}[v_1(x',t',c,\varepsilon_1{}')|(m)] &= \int_c \int_{t'} \int_{x'} \sigma \log \left(\sum_{d_1{}' \in \{T,R\}} e^{u_1(x',t',c,d_1{}')/\sigma} \right) dF_x(x') dF_t(t') dF_{c|(m)}(c|(m)) \\ &= \int_c \int_{t_1{}'} \cdots \int_{t_{J_t}{}'} \int_{x_1{}'} \cdots \int_{x_{J_x}{}'} \left[\cdot \right] dF_{x,1}(x_1{}'|x_2', \ldots, x_{J_x}{}') \cdots dF_{x,J_x}(x_{J_x}{}') \\ &\quad dF_{t,1}(t_1{}'|t_2', \ldots, t_{J_t}{}') \cdots dF_{t,J_t}(t_{J_t}{}') dF_{c|(m)}(c|(m)). \end{aligned}$$

As such, Round 2's conditional choice probabilities are:

$$P(d_2|x,z,t,m,c_{-1}) = \frac{\exp(u_2(x,z,t,m,c_{-1},d_2)/\sigma)}{\sum_{\widetilde{d_2}\in\{A,R\}}\exp(u_2(x,z,t,m,c_{-1},\widetilde{d_2})/\sigma)}. \tag{20}$$

Each year's Round 1 value function is:

$$v_1(x,t,c_{-1},\varepsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,c_{-1},d_1)+\varepsilon_1(d_1), \text{ where:} \tag{21}$$

$$\begin{aligned} u_1(x,t,c_{-1},d_1) &= \mathbb{E}[v_2(x,z,t,m,c_{-1},\varepsilon_2)|x,t,c_{-1}]\mathbb{1}(d_1=T) \\ &\quad + \{L(y_1^*=S|t)+\beta\mathbb{E}[v_1(x',t',c,\varepsilon_1{}')]\}\mathbb{1}(d_1=R), \text{ where:} \end{aligned}$$

$$\begin{aligned} &\mathbb{E}[v_2(x,z,t,m,c_{-1},\varepsilon_2)|x,t,c_{-1}] \\ &\quad = \int_{\widetilde{m}}\int_{\widetilde{z}}\sigma\log\left(\sum_{\widetilde{d_2}\in\{A,R\}} e^{u_2(x,\widetilde{z},t,\widetilde{m},c_{-1},\widetilde{d_2})/\sigma}\right)dF_{z|x,t}(\widetilde{z}|x,t)dF_{m|x,t}(\widetilde{m}|x,t) \\ &\quad = \int_{\widetilde{m}}\int_{\widetilde{z_1}}\cdots\int_{\widetilde{z_{J_z}}}\left[\cdot\right]dF_{z|x,t,1}(\widetilde{z_1}|\widetilde{z_2},\ldots,\widetilde{z_{J_z}},x,t)\cdots dF_{z|x,t,J_z}(\widetilde{z_{J_z}}|x,t)dF_{m|x,t}(\widetilde{m}|x,t). \end{aligned}$$

And as such, Round 1's conditional choice probabilities (CCPs) are:

$$P(d_1|x,t,c_{-1}) = \frac{\exp(u_1(x,t,c_{-1},d_1)/\sigma)}{\sum_{\widetilde{d_1}\in\{T,R\}}\exp(u_1(x,t,c_{-1},\widetilde{d_1})/\sigma)}. \tag{22}$$

The unrestricted CCPs, success probabilities, and cutoff levels are $P_{d_1} = P\big(d_1 = T\big|x,t,c_{-1};\theta_T\big)$, $P_{d_2} = P\big(d_2 = A\big|x,z,m,c_{-1};\theta_A\big)$, $P_y = P\big(y = S\big|x,z,t;\theta_S\big)$, $L_{y_1^*} = L\big(y_1^* = S\big|t;\theta_{S_1^*}\big)$, and $L_{y_2^*} = L\big(y_2^* = S\big|c_{-1};\theta_{S_2^*}\big)$, which aside from the cutoffs have the usual logistic specification. In this model, the cutoffs are not in a function that restricts their range. Given that, the unrestricted likelihood is:

$$\begin{aligned} \mathcal{L}_N^U(\theta) &= \prod_{n=1}^{N_A} f\big(c_n|m_n;\theta_{F_c}\big)\prod_{n=1}^{N_T} f\big(z_n^c|z_n^{\neg c},x_n,t_n;\theta_{F_z}\big)f\big(m_n|x_n,t_n;\theta_{F_m}\big) \\ &\quad P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}\prod_{n=N_T+1}^{N}\big(1-P_{d_{1,n}}(\theta_T)\big) \\ &\quad \prod_{n=1}^{N} f\big(x_n;\theta_{F_x}\big)f\big(t_n;\theta_{F_t}\big)f\big(z_n^{\neg c}|x_n,t_n;\theta_{F_z}\big)P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}. \end{aligned} \tag{23}$$

As with the dynamic models in the paper, I use (θ_F^U,θ_S^U) as an initial guess of (θ_F^R,θ_S^R). To show that the dynamics of this model work as intended, I prove the following theorem:

Theorem 2. *Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. Suppose there are two arbitrary applicants in year t indexed by H and L, such that $P(y = S|x^H, z^H, t) = P(y = S|x^L, z^L, t)$, but $m^H > m^L$. That is, H has the higher matriculation probability. I will show that $P(d_2 = A|x^H, z^H, t, m^H, c_{-1}) > P(d_2 = A|x^L, z^L, t, m^L, c_{-1})$.

Define for each round the choice-specific payoffs for acceptance (A) and rejection (R):

$$\begin{aligned} \pi_2^A &\equiv P(y = S|x, z, t) + \beta \mathbb{E}[v_1(x', t', c, \varepsilon_1')|m], \quad \pi_2^R \equiv L(y_2^* = S|c_{-1}) + \beta \mathbb{E}[v_1(x', t', c, \varepsilon_1')], \\ \pi_1^A &\equiv \mathbb{E}[v_2(x', z', t', m', c, \varepsilon_2')|x', t', c], \quad \pi_1^R \equiv L(y_1^{*\prime} = S|t') + \beta \mathbb{E}[v_1(x'', t'', c', \varepsilon_1'')]. \end{aligned} \tag{24}$$

It is the case that:

$$P(d_2 = A|x, z, t, m, c_{-1}) = \frac{\exp(\pi_2^A/\sigma)}{\exp(\pi_2^A/\sigma) + \exp(\pi_2^R/\sigma)}, \text{ where:} \tag{25}$$

$$\mathbb{E}[v_1(x', t', c, \varepsilon_1')|m] = \int_c \int_{t'} \int_{x'} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(x')\, dF_t(t')\, dF_{c|m}(c|m), \text{ where:}$$

$$\mathbb{E}[v_2(x', z', t', m', c, \varepsilon_2')|x', t', c] = \int_{\widetilde{m'}} \int_{\widetilde{z'}} \sigma \log\left(e^{\pi_2^{A\prime}/\sigma} + e^{\pi_2^{R\prime}/\sigma}\right) dF_{z|x,t}(\widetilde{z'}|x', t') dF_{m|x,t}(\widetilde{m'}|x', t'),$$

where $\pi_2^{A\prime}$ and $\pi_2^{R\prime}$ are defined analogously to Equation (24) at the next year's states.

Per-period utilities are bounded and $\beta \in (0, 1)$, so the value functions exist and are unique. The only place m appears in $P(d_2 = A|x, z, t, m, c_{-1})$ is in the distribution of c conditional on m. I parameterize $c|m \sim \mathcal{N}(m\beta_m, \sigma_m^2)$, where by assumption $\beta_m > 0$. β_m is positive because higher matriculation probabilities lead to a greater expected cohort size. Multiple years of data are necessary to identify β_m, since cohort sizes vary only across years. Then:

$$F_{c|m}(c|m^H; \beta_m, \sigma_m^2) = \Phi\left(\frac{c - m^H\beta_m}{\sigma_m}\right) < \Phi\left(\frac{c - m^L\beta_m}{\sigma_m}\right) = F_{c|m}(c|m^L; \beta_m, \sigma_m^2). \tag{26}$$

Therefore, $c|m^H$ first-order stochastically dominates $c|m^L$. To apply this stochastic dominance, I must show that $\mathbb{E}[v_1(x', t', c, \varepsilon_1')|m]$'s integrand consisting of all but the outermost integral is strictly increasing in c. But the only place c appears there is in $\mathbb{E}[v_2(x', z', t', m', c, \varepsilon_2')|x', t', c]$, and the only place c appears in that expectation is in $L(y_2^{*\prime} = S|c; \theta_{S_2^*}) = f_{S_2^*}(c)\theta_{S_2^*}$.

By assumption, $f_{S_2^*}(\cdot)$ is strictly increasing and $\theta_{S_2^*} > 0$. $\theta_{S_2^*, c}$ is positive because a greater cohort size this year means that the committee, needing to keep the total number of students in the program relatively constant across years, can afford to be more selective next year. As such, I have that $P(y_2^{*\prime} = S|c)$, and in turn $\mathbb{E}[v_1(x', t', c, \varepsilon_1')|m]$'s integrand, are strictly increasing in c. Combining this result with $[c|m^H] \succ_{FOSD} [c|m^L]$, I obtain the inequality:

$$\begin{aligned} \mathbb{E}[v_1(x', t', c, \varepsilon_1')|m^H] &= \int_c \int_{t'} \int_{x'} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(x')\, dF_t(t')\, dF_{c|m}(c|m^H) \\ &> \int_c \int_{t'} \int_{x'} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(x')\, dF_t(t')\, dF_{c|m}(c|m^L) \\ &= \mathbb{E}[v_1(x', t', c, \varepsilon_1')|m^L]. \end{aligned} \tag{27}$$

Therefore, by Equation (25), Inequality (27), and $P(y=S|x^H,z^H,t)=P(y=S|x^L,z^L,t)$, I have that $P(d_2=A|x^H,z^H,t,m^H,c_{-1})>P(d_2=A|x^L,z^L,t,m^L,c_{-1})$. Intuitively, if the committee rejects either applicant, their m does not matter. But since their success probabilities are equal, accepting applicant H yields a higher expected payoff than accepting applicant L. □

D.2 Diversity Model

I remove the state variable c_{-1} and add in its place two new states a and $\overline{a}_{-1}$. Let $a=\mathbb{1}(\text{Diverse})$, where a diverse characteristic is one whose probability of appearing among next year's applicants is bounded above by some $p\in(0,1)$. For example, I could have $a=\mathbb{1}(\text{TheoryField})$, where in most years, theory applicants are less common than applied applicants. Also, let $\overline{a}_{-1}$ equal the percentage of last year's acceptance set that is theory. If $|a-\overline{a}_{-1}|$ shrinks, then the applicant's expected success probability decreases, as their future advisor may have less time for them.[9]

As such, the optimal dynamic decision may not match the optimal myopic one. In particular, the committee may accept a diverse applicant it would not have otherwise because doing so increases the success probability of next year's applicant pool. Accepting a diverse applicant increases the distribution of $\overline{a}$ conditional on a, which because diverse applicants are less common increases the distribution of $|a'-\overline{a}|$, which increases next year's expected success probability, which increases next year's expected value. That said, I can also estimate a myopic model with memory variable $\overline{a}_{-1}$. The committee will still prioritize diverse applicants because diversity increases applicants' success probabilities, but there will be no forward-looking dynamic effect.

I now present the value functions, starting with each year's Round 2 value:

$$v_2(x,z,t,a,\overline{a}_{-1},\varepsilon_2)=\max_{d_2\in\{A,R\}}u_2(x,z,t,a,\overline{a}_{-1},d_2)+\varepsilon_2(d_2),\text{ where:} \tag{28}$$

$$\begin{aligned}u_2(x,z,t,a,\overline{a}_{-1},d_2)=&\{P(y=S|x,z,t,|a-\overline{a}_{-1}|)+\beta\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')|a]\}\mathbb{1}(d_2=A)\\&+\{L(y_2^*=S|t)+\beta\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')]\}\mathbb{1}(d_2=R),\text{ where:}\end{aligned}$$

$$\begin{aligned}\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')|a]&=\int_{\widetilde{\overline{a}}}\int_{\widetilde{t'}}\int_{\widetilde{x'}}\sigma\log\left(\sum_{d_1'\in\{T,R\}}e^{u_1(\widetilde{x'},\widetilde{t'},\widetilde{\overline{a}},d_1')/\sigma}\right)dF_{x'}(\widetilde{x'})dF_{t'}(\widetilde{t'})dF_{\overline{a}|a}(\widetilde{\overline{a}}|a)\\&=\int_{\widetilde{\overline{a}}}\int_{\widetilde{t'_1}}\cdots\int_{\widetilde{t'_{J_t}}}\int_{\widetilde{x'_1}}\cdots\int_{\widetilde{x'_{J_x}}}\left[\cdot\right]dF_{x,1}(x_1{}'|x_2{}',\ldots,x_{J_x}{}')\cdots dF_{x,J_x}(x_{J_x}{}')\\&\quad dF_{t,1}(t_1{}'|t_2{}',\ldots,t_{J_t}{}')\cdots dF_{t,J_t}(t_{J_t}{}')dF_{\overline{a}|a}(\widetilde{\overline{a}}|a).\end{aligned}$$

As such, Round 2's conditional choice probabilities are:

$$P(d_2|x,z,t,a,\overline{a}_{-1})=\frac{\exp(u_2(x,z,t,a,\overline{a}_{-1},d_2)/\sigma)}{\sum_{\widetilde{d_2}\in\{A,R\}}\exp(u_2(x,z,t,a,\overline{a}_{-1},\widetilde{d_2})/\sigma)}. \tag{29}$$

[9] If the program never gets too imbalanced, this relationship will not be identifiable, in which case estimation will not work for this choice of a. However, there may also be different suitable diversity variables.

Each year's Round 1 value function is:

$$v_1(x,t,\overline{a}_{-1},\varepsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,\overline{a}_{-1},d_1) + \varepsilon_1(d_1), \text{ where:} \tag{30}$$

$$\begin{aligned} u_1(x,t,\overline{a}_{-1},d_1) &= \mathbb{E}[v_2(x,z,t,a,\overline{a}_{-1},\varepsilon_2)|x,t,\overline{a}_{-1}]\mathbb{1}(d_1=T) \\ &\quad + \{L(y_1^*=S|t) + \beta\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')]\}\mathbb{1}(d_1=R), \text{ where:} \end{aligned}$$

$$\begin{aligned} \mathbb{E}[v_2(x,z,t,a,\overline{a}_{-1},\varepsilon_2)|x,t,\overline{a}_{-1}] &= \int_{\widetilde{a}}\int_{\widetilde{z}} \sigma\log\left(\sum_{\widetilde{d}_2\in\{A,R\}} e^{u_2(x,\widetilde{z},t,\widetilde{a},\overline{a}_{-1},\widetilde{d}_2)/\sigma}\right) dF_{z|x,t}(\widetilde{z}|x,t)dF_a(\widetilde{a}) \\ &= \int_{\widetilde{a}}\int_{\widetilde{z}_1}\cdots\int_{\widetilde{z}_J}\left[\cdot\right] dF_1(\widetilde{z}_1|\widetilde{z}_2,\ldots\widetilde{z}_J,x,t)\cdots dF_J(\widetilde{z}_J|x,t)dF_a(\widetilde{a}) \end{aligned}$$

$$\begin{aligned} \mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')] &= \int_{\widetilde{a}}\int_{\widetilde{t'}}\int_{\widetilde{x'}} \sigma\log\left(\sum_{d_1'\in\{T,R\}} e^{u_1(\widetilde{x'},\widetilde{t'},\widetilde{a},d_1')/\sigma}\right) dF_{x'}(\widetilde{x'})dF_{t'}(\widetilde{t'})dF_{\overline{a}}(\widetilde{a}) \\ &= \int_{\widetilde{a}}\int_{\widetilde{t_1'}}\cdots\int_{\widetilde{t_{J_t}'}}\int_{\widetilde{x_1'}}\cdots\int_{\widetilde{x_{J_x}'}}\left[\cdot\right] dF_{x,1}(x_1'|x_2',\ldots,x_{J_x}')\cdots dF_{x,J_x}(x_{J_x}') \\ &\quad dF_{t,1}(t_1'|t_2',\ldots,t_{J_t}')\cdots dF_{t,J_t}(t_{J_t}')dF_{\overline{a}}(\widetilde{a}). \end{aligned}$$

And as such, Round 1's conditional choice probabilities are:

$$P(d_1|x,t,\overline{a}_{-1}) = \frac{\exp(u_1(x,t,\overline{a}_{-1},d_1)/\sigma)}{\sum_{\widetilde{d}_1\in\{T,R\}}\exp(u_1(x,t,\overline{a}_{-1},\widetilde{d}_1)/\sigma)}. \tag{31}$$

The unrestricted CCPs, success probabilities, and cutoff levels are $P_{d_1} = P\big(d_1 = T\big|x,t,\overline{a}_{-1};\theta_T\big)$, $P_{d_2} = P\big(d_2 = A\big|x,z,t,a,\overline{a}_{-1};\theta_A\big)$, $P_y = P\big(y = S\big|x,z,t,|a-\overline{a}_{-1}|;\theta_S\big)$, $L_{y_1^*} = L\big(y_1^* = S\big|t;\theta_{S_1^*}\big)$, and $L_{y_2^*} = L\big(y_2^* = S\big|t;\theta_{S_2^*}\big)$; all but the cutoffs are logistic. Given that, the unrestricted likelihood is:

$$\begin{aligned} \mathcal{L}_N^U(\theta) &= \prod_{n=1}^{N_A} f\big(\overline{a}_n|a_n;\theta_{F_{\overline{a}}}\big)\prod_{n=1}^{N_T} f\big(z_n^c|z_n^{\neg c},x_n,t_n;\theta_{F_z}\big)f\big(a_n;\theta_{F_a}\big) \\ &\quad P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}\prod_{n=N_T+1}^{N}\big(1-P_{d_{1,n}}(\theta_T)\big) \\ &\quad \prod_{n=1}^{N} f\big(x_n;\theta_{F_x}\big)f\big(t_n;\theta_{F_t}\big)f\big(z_n^{\neg c}|x_n,t_n;\theta_{F_z}\big)P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}. \end{aligned} \tag{32}$$

I estimate $f_{\overline{a}|a}(\overline{a}|a)$ on the N_A accepted applicants only (those from N_A+1 to $N_{T,R}$ are rejected in Round 2) because only accepted applicants can influence the acceptance set's area of interest. Also, as with the dynamic models in the paper, I use (θ_F^U,θ_S^U) as an initial guess of (θ_F^R,θ_S^R). To show that the dynamics of this model work as intended, I prove the following theorem:

Theorem 3. *Consider any two applicants, one with $a=1$ and one with $a=0$, but with equal success probabilities and therefore different values of x or z. Suppose $P(a'=1)=p>0$. Then for all sufficiently small p, the $a=1$ applicant has a strictly greater Round 2 acceptance probability.*

Proof. Suppose there are two arbitrary applicants in year t, one with $a = 1$ and the other with $a = 0$, but with $P(y = S|x^1, z^1, t, 1 - \overline{a}_{-1}) = P(y = S|x^0, z^0, t, \overline{a}_{-1})$. Unlike in the previous two proofs, these equal success probabilities imply that unless $\overline{a}_{-1} = \frac{1}{2}$, it must be that $(x^1, z^1) \neq (x^0, z^0)$. I will show that $P(d_2 = A|x^1, z^1, t, 1, \overline{a}_{-1}) > P(d_2 = A|x^0, z^0, t, 0, \overline{a}_{-1})$.

Define for each round the choice-specific payoffs for acceptance (A) and rejection (R):

$$\pi_2^A \equiv P(y = S|x, z, t, |a - \overline{a}_{-1}|) + \beta \mathbb{E}[v_1(x', t', \overline{a}, \varepsilon_1')|a], \quad \pi_2^R \equiv L(y_2^* = S|t) + \beta \mathbb{E}[v_1(x', t', \overline{a}, \varepsilon_1')],$$

$$\pi_1^A \equiv \mathbb{E}[v_2(x', z', t', a', \overline{a}, \varepsilon_2')|x', t', \overline{a}], \quad \pi_1^R \equiv L(y_1^{*\prime} = S|t') + \beta \mathbb{E}[v_1(x'', t'', \overline{a}', \varepsilon_1'')]. \tag{33}$$

It is the case that:

$$P(d_2 = A|x, z, t, a, \overline{a}_{-1}) = \frac{\exp(\pi_2^A/\sigma)}{\exp(\pi_2^A/\sigma) + \exp(\pi_2^R/\sigma)}, \text{ where:}$$

$$\mathbb{E}[v_1(x', t', \overline{a}, \varepsilon_1')|a] = \int_{\widetilde{\overline{a}}} \int_{\widetilde{t'}} \int_{\widetilde{x'}} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(\widetilde{x'}) \, dF_t(\widetilde{t'}) \, dF_{\overline{a}|a}(\widetilde{\overline{a}}|a), \text{ where:}$$

$$\begin{aligned}\mathbb{E}[v_2(x', z', t', a', \overline{a}, \varepsilon_2')|x', t', \overline{a}] = P(a' = 1) \int_{\widetilde{z'}} \sigma \log\left(e^{\pi_2^{A\prime}(1)/\sigma} + e^{\pi_2^{R\prime}/\sigma}\right) dF_{z|x,t}(\widetilde{z'}|x', t') \\ + [1 - P(a' = 1)] \int_{\widetilde{z'}} \sigma \log\left(e^{\pi_2^{A\prime}(0)/\sigma} + e^{\pi_2^{R\prime}/\sigma}\right) dF_{z|x,t}(\widetilde{z'}|x', t'),\end{aligned} \tag{34}$$

where $\pi_2^{A\prime}(a')$ denotes next year's Round 2 acceptance payoff evaluated at $a' \in \{0, 1\}$.

Per-period utilities are bounded and $\beta \in (0, 1)$, so the value functions exist and are unique. The only two places a appears in $P(d_2 = A|x, z, t, a, \overline{a}_{-1})$ are in the success probabilities, which are assumed to be equal for the $a = 1$ and $a = 0$ applicants, and in the distribution of $\overline{a}$ conditional on a. I parameterize $\overline{a}|a \sim \mathcal{N}(a\beta_a, \sigma_a^2)$, where by assumption $\beta_a > 0$. β_a is positive because $a = 1$ applicants are more likely than $a = 0$ applicants to be in a year with a high percentage of accepted $a = 1$ applicants. As such, multiple years of data are necessary to identify β_a. Then:

$$F_{\overline{a}|a}(\overline{a}|1; \beta_a, \sigma_a^2) = \Phi\left(\frac{\overline{a} - \beta_a}{\sigma_a}\right) < \Phi\left(\frac{\overline{a}}{\sigma_a}\right) = F_{\overline{a}|a}(\overline{a}|0; \beta_a, \sigma_a^2). \tag{35}$$

Therefore, $\overline{a}|1$ first-order stochastically dominates $\overline{a}|0$. To apply this stochastic dominance, I must show that $\mathbb{E}[v_1(x', t', \overline{a}, \varepsilon_1')|a]$'s integrand consisting of all but the outermost integral is strictly increasing in $\overline{a}$. But the only place $\overline{a}$ appears there is in $\mathbb{E}[v_2(x', z', t', a', \overline{a}, \varepsilon_2')|x', t', \overline{a}]$, and the only place $\overline{a}$ appears in that expectation is in $P(y' = S|x', z', t', \overline{a}; \theta_S) = \frac{\exp(f_S(x', z', t', |a' - \overline{a}|)\theta_S)}{1 + \exp(f_S(x', z', t', |a' - \overline{a}|)\theta_S)}$.

By assumption, $f_S(\cdot)$ is strictly increasing and $\theta_{S,|a-\overline{a}_{-1}|} > 0$. $\theta_{S,|a-\overline{a}_{-1}|}$ is positive because a large value of $|a - \overline{a}_{-1}|$ means that applicant a is diverse relative to the previous year's accepted applicants and is therefore more likely to succeed. Recall $P(a' = 1) = p$. Taking the limit $p \to 0$:

$$\begin{aligned}\mathbb{E}[v_2(x', z', t', a', \overline{a}, \varepsilon_2')|x', t', \overline{a}] \to \int_{\widetilde{z'}} \sigma \log(e^{\{P(y'=S|x', z', t', \overline{a}) + \beta \mathbb{E}[v_1(x'', t'', \overline{a}', \varepsilon_1'')|0]\}/\sigma} \\ + e^{\{L(y_2^{*\prime} = S|t') + \beta \mathbb{E}[v_1(x'', t'', \overline{a}', \varepsilon_1'')]\}/\sigma}) dF_{z|x,t}(\widetilde{z'}|x', t').\end{aligned} \tag{36}$$

$P(y' = S|x', z', t', \overline{a})$ is strictly increasing in $\overline{a}$, and by the continuity of $\mathbb{E}[v_2(x', z', t', a', \overline{a}, \varepsilon_2')|$ $x', t', \overline{a}]$ in p, I have that it, and in turn $\mathbb{E}[v_1(x', t', \overline{a}, \varepsilon_1')|a]$'s integrand, are also strictly increasing in $\overline{a}$ for sufficiently small p. Combining this result with $\overline{a}|1 \succ_{FOSD} \overline{a}|0$, I obtain the inequality:

$$\begin{aligned}
\mathbb{E}[v_1(x', t', \overline{a}, \varepsilon_1')|1] &= \int_{\widetilde{\overline{a}}} \int_{\widetilde{t'}} \int_{\widetilde{x'}} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(\widetilde{x'})\, dF_t(\widetilde{t'})\, dF_{\overline{a}|a}(\widetilde{\overline{a}}|1) \\
&> \int_{\widetilde{\overline{a}}} \int_{\widetilde{t'}} \int_{\widetilde{x'}} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(\widetilde{x'})\, dF_t(\widetilde{t'})\, dF_{\overline{a}|a}(\widetilde{\overline{a}}|0) \qquad (37) \\
&= \mathbb{E}[v_1(x', t', \overline{a}, \varepsilon_1')|0].
\end{aligned}$$

Therefore, by Equation (34), Inequality (37), and $P(y = S|x^1, z^1, t, 1 - \overline{a}_{-1}) = P(y = S|x^0, z^0, t, \overline{a}_{-1})$, I have $P(d_2 = A|x^1, z^1, t, 1, \overline{a}_{-1}) > P(d_2 = A|x^0, z^0, t, 0, \overline{a}_{-1})$. Intuitively, if the committee rejects either applicant, their a does not matter. But since their success probabilities are equal, accepting applicant $a = 1$ yields a higher expected payoff than accepting $a = 0$.

In this proof, I must assume $P(a' = 1)$ is sufficiently small, and not just $P(a' = 1) < \frac{1}{2}$. The reason is that if $P(a' = 1)$ is too large, then $\mathbb{E}[v_2(x', z', t', a', \overline{a}, \varepsilon_2')|x', t', \overline{a}]$ may not be strictly increasing in $\overline{a}$, considering $P(y' = S|x', z', t', 1 - \overline{a})$ is strictly decreasing in $\overline{a}$. This indeterminacy can be explored by calculating $\frac{\partial}{\partial \overline{a}} \mathbb{E}[v_2(x', z', t', a', \overline{a}, \varepsilon_2')|x', t', \overline{a}]$. As such, to be sure the committee is more likely to accept the $a = 1$ applicant, the committee must know that there will be sufficiently few $a' = 1$ applicants. This assumption is strong, but possible if $a' = 1$ is sufficiently rare for the committee to know it will happen. But even without forward-looking dynamics, if $\overline{a}_{-1} < \frac{1}{2}$, then the committee will be more likely to accept an $a = 1$ applicant than an $a = 0$ applicant with identical x and z values. The reason is that $P(y = S|x, z, t, 1 - \overline{a}_{-1}) > P(y = S|x, z, t, \overline{a}_{-1})$. □

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